

PORTFOLIO TERM 1/2

ENVT-1101-M01-2024-25-130

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TERM 1



Canvey Commons

ENVT-1101-M01-2024-25 Landscape Design 2:

Ecologies, Exploration and Proposition

Khaled Mohamed-MLA1

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Future metabolism of sandy bays prefab suburbia

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Current site: site and section

Canvey -2100

Canvey -2050

Canvey -2075

Section analysis

Veiw -1

Veiw -2

Veiw -3



Exploring Canvey

Canvey Island is a manmade landscape of adaptation. Located in the Thames Estuary, much of the land lies below sea level, held in place by a 14 mile sea wall constructed after the catastrophic 1953 North Sea flood, which claimed 58 lives on the island and displaced 13,000 people.

Before that, the 1731 and 1897 floods similarly reshaped land policy and the physical form of the island. Flooding is not an event here; it is a pattern.

The current sea wall protects 38,000 residents and critical infrastructure (Environment Agency, 2021). It is a barrier, but historically it was more: a social edge, a public balcony facing the estuary, a communal path.



Current Canvey map



Canvey 2100 prediction

The Wall

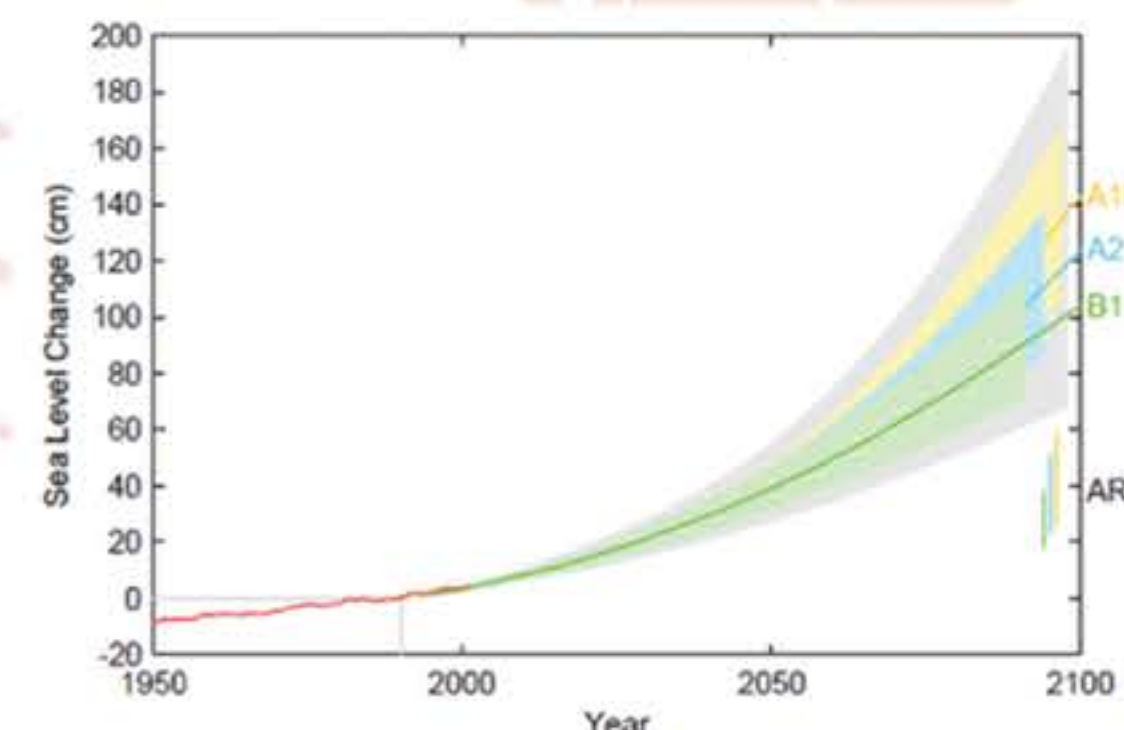
Canvey's concrete sea wall is the only thing separating the town from tidal flooding. It was constructed after the 1953 North Sea Flood and reinforced in the 1980s. While it currently protects over 38,000 residents, it is finite in both height and lifespan.

The wall is not going anywhere now. It is too large to dismantle, too structurally embedded to relocate, and too necessary to ignore. But that does not make it sacred.

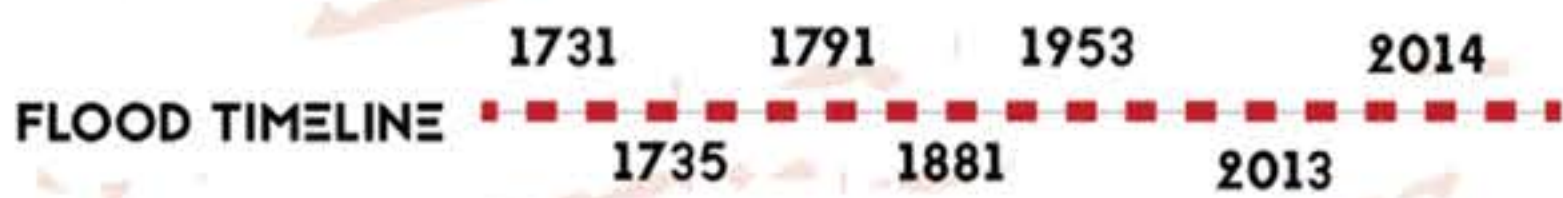
This project does not glorify concrete. It integrates it. It sees the wall not as a final gesture, but as a starting datum, a ridge to build from, bury into, and reclaim as terrain.



Source: Canvey Community Archive



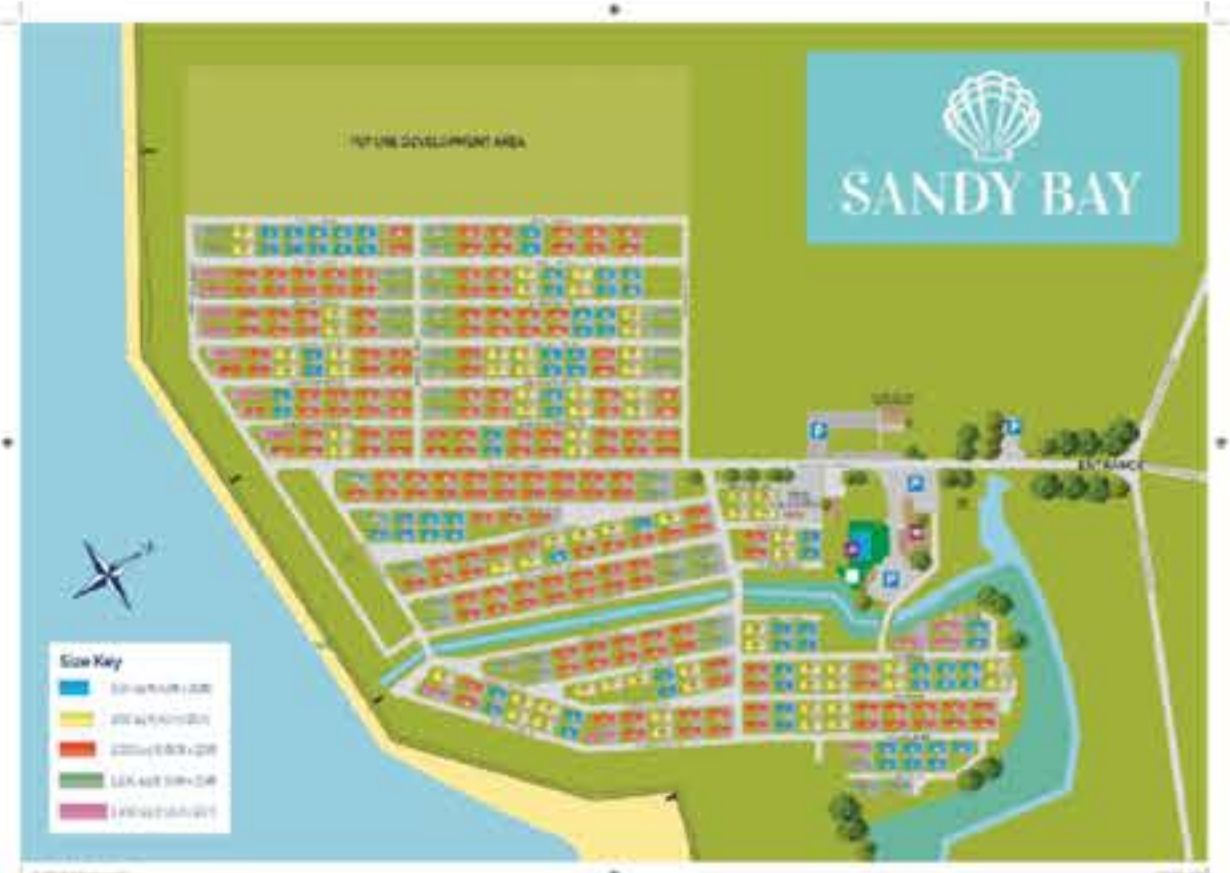
Source: Stefan Rahmstorf, PIK (2012), Nature Education Knowledge 3(10):4



Thorney Bay as "Sandy Bay"

Thorney Bay, now rebranded as "Sandy Bay," has long been a liminal zone on Canvey Island – a place between land and water, permanence and escape. In the mid-20th century, it operated as a caravan park and informal beach camp, attracting seasonal visitors and temporary residents who set up close to the sea.

This informality was erased in the early 2000s, when the site was converted into a gated housing enclave made up of factory-built park homes. It was renamed Sandy Bay and marketed to over-50s seeking peace, affordability, and proximity to the coast. The beach was fenced. Access was restricted. The name changed. The typology changed. The public edge became private real estate.



Thorney bay beach camp 1955-1960

Historical archives show that the area of Thorney Bay was once much less restricted, serving as a place where people enjoyed themselves and engaged in their own unique ways. This is evident in the varied use of space and the diversity of building and dwelling styles.

Engaging as an active element of the landscape, the Canvey Sea Wall has historically been a vibrant part of community life, used and interacted with in a variety of ways. This structure has not only provided protection but also served as a gathering space, a vantage point, and a place for recreation, reflecting the community's close relationship with the landscape.



Today, Sandy Bay is a private, fenced housing estate composed almost entirely of prefabricated park homes manufactured off-site. These single-storey units are sold as "virtual freeholds," meaning the buyer owns the home, but not the land – which is leased from the estate operator.

The area is spatially disconnected from the rest of Canvey Island. It is bordered on one side by the sea wall and on the other by fences and car-dominated infrastructure. There are no schools and no shops inside the gated area. It is a landscape of final residency.

NOW



onthemarket.com

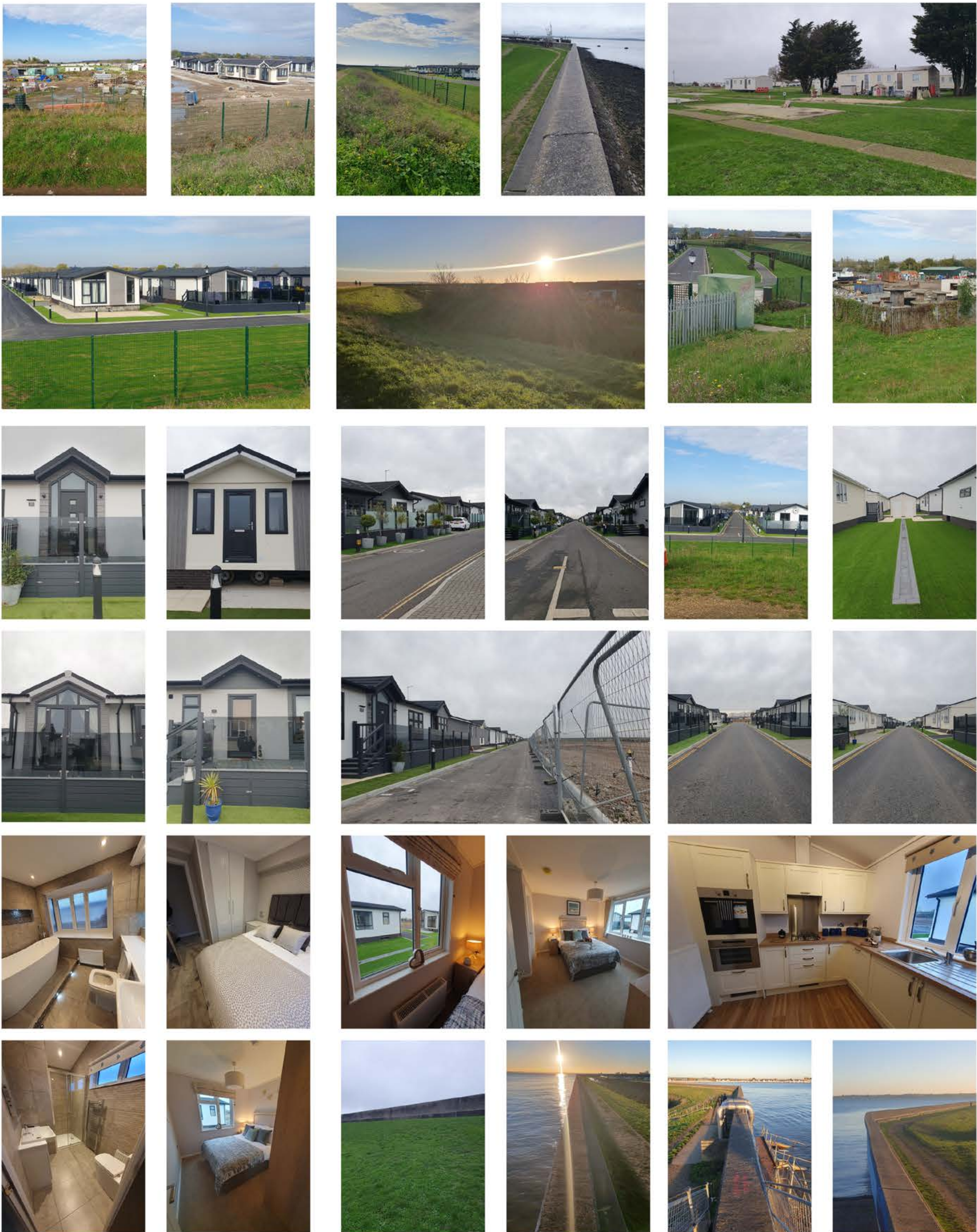
onthemarket.com

Site Visit

Typology & Edges:

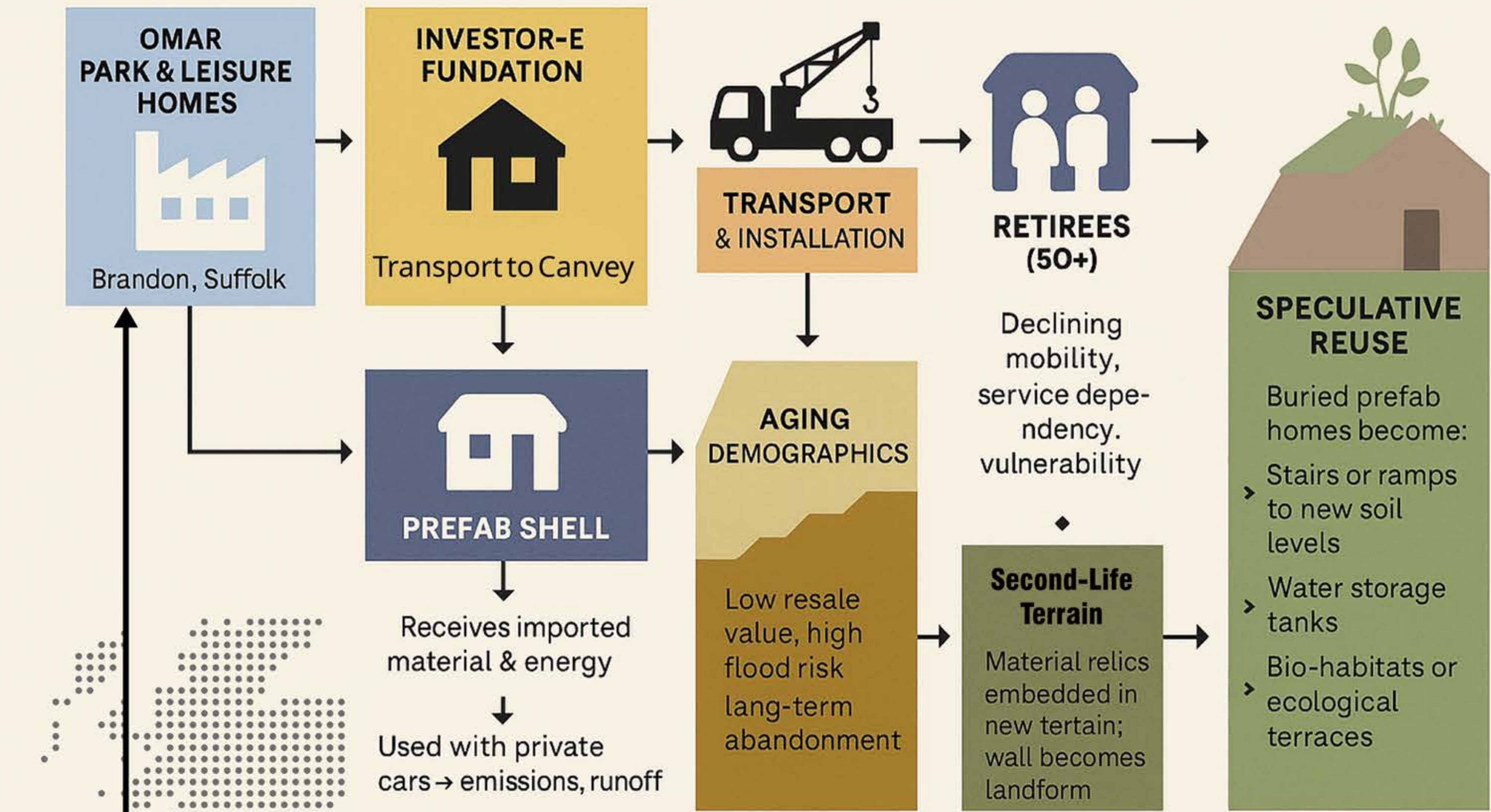
The houses follow a fixed model: single-storey pre-fab units, installed on shallow pads with minor aesthetic variation. Exteriors are clean but fenced, with artificial turf and planters used to soften repetition. Internally, rooms are compact, domestic, and softly furnished – but reveal light construction, limited insulation, and minimal depth.

At the site scale, the edge conditions are controlled and uniform. Roads are wide and central, while boundaries are hard: fence, kerb, sea wall. The wall itself remains unengaged – physically close, but spatially and architecturally ignored.



FUTURE METABOLISM OF SANDY BAY'S PREFAB SUBURBIA

Deconstructing and Regrounding the Housing of 2100



pre-designed, prefabricated unit manufactured in Brandon, Suffolk, craned onto leased floodplain plots on Canvey Island.

INPUTS

Flow Type	Description	Source
Electricity	Power for heating, cooking, lighting	National Grid (fossil mix)
Water	Domestic use, sanitation, occasional irrigation	Municipal mains
Materials	Bitumen, vinyl, insulation, plasterboard	Omar factory (Brandon, Suffolk)
Mobility	Private cars parked onsite	Fossil fuel + embodied energy
Logistics	Prefab delivery + crane operation	Trucked from manufacturing hub

In-House Processes

Function	Description	Notes
Heating	Space + water heating (usually electric or LPG)	Inefficient; low thermal mass
Domestic Use	Washing, food prep, air conditioning (optional)	Standard UK domestic consumption
Occupancy	Over-50s residents, seasonal or permanent	Demographic input + aging
Appliance Load	TV, fridge, cooker, HVAC	Medium load with spikes in winter

OUTPUTS

Output Type	Description	Notes
Heat Loss	Through lightweight envelope	Poor insulation lifespan
Grey + Blackwater	Water waste from domestic use	Municipal sewer discharge
Solid Waste	Packaging, household waste	Standard bin collection
Material Aging	Roof deterioration, plastic embrittlement	Non-circular, destined for landfill
Social Emission	Vacancy, health dependency, isolation	Adds to service burden on island

1. Roof Finish
Material: Bitumen shingles or synthetic slate
Function: Waterproofing and heat shedding
Life Expectancy: 20–30 years (subject to UV and coastal exposure)

2100 Speculation: Warped, moss-covered shell; could serve as canopy or shade for post-flood access ramps.

2. Roof Truss Structure
Material: Lightweight softwood, prefabricated
Function: Supports roof loads, enables modular delivery
Life Expectancy: 30–40 years with maintenance; risk of rot if exposed

2100 Speculation: May be salvaged as temporary framing for community structures or stair access embedded in raised soil banks.

3. Wall Panels
Material: Timber stud frame with rigid insulation and vinyl/PVC cladding
Function: Thermal envelope and weather barrier
Life Expectancy: 25–40 years (vinyl subject to UV degradation)

2100 Speculation: Outer skin becomes brittle; frame may survive burial, forming cavities or reused as infiltration chambers.

4. Floor Structure
Material: Insulated timber joists over steel or timber base chassis
Function: Supports internal loading and raises unit above ground
Life Expectancy: 30–50 years (dependent on moisture exposure)

2100 Speculation: May partially collapse; could be repurposed as water storage basin or shallow substructure under raised terrain.

By 2100, its metabolism collapses. Yet in that decay lies potential: its parts may scaffold a new terrain, a different ecology, a future metabolic logic.

Using renewable energy sources such as hydroelectric power, built directly into the walls.	Resilient structures to rising sea levels through Adaptation and scalable design.	Foster inclusiveness by designing modular housing accessible to all economic levels.
Encourage biodiversity by integrating vegetation and habitats into architectural designs.	Purify watering systems using some of natural systems' alternative methods of filtration with vegetation.	Design common space as a mix of industry, living, and leisure use, which acts as a community hub.
Close socio-economic gaps through community integration created by shared spaces.	Maximize efficiency through recycling materials and reusing leftover waste.	Create adaptive, self-sustaining infrastructures for long-term climate resilience.
Establish safe and secure living conditions by incorporating active surveillance and lighting.	Encourage local economies by supporting fishing activities and open markets.	Preserve cultural identity by reflecting Canvey Island's historical narratives in design.
Integrate educational elements such as learning hubs, to foster awareness of sustainability.	Use eco-friendly materials to minimize carbon footprint and environmental degradation.	Adaptive Urban Layers through multi-functional wall integrating housing, ecology, and commerce.

Canvey Commons

Spatial Rules

1



No fencing

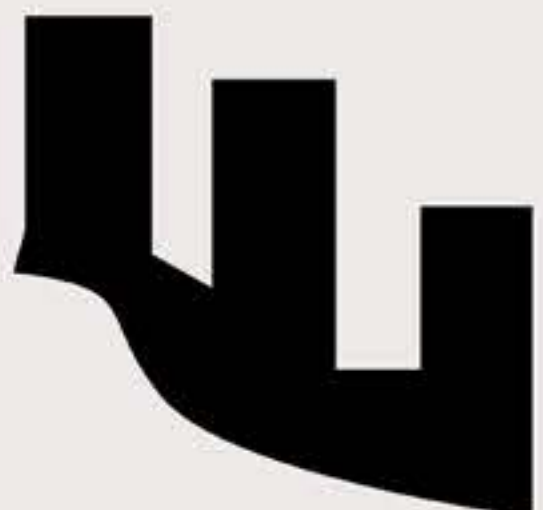
Applies to: All civic and shared edge zones

Ban: No metal fences between units or paths

Replacement: Hedgerows, planters, civic walls

Goal: Dismantle visual and spatial privatization

2



Buried Housing

Applies to: Obsolete prefab units

Action: Bury structures in soil slope (min 1m cover)

Function: Become landform base, not waste

Surface use: Seating, terraces, stairs, gardens

3



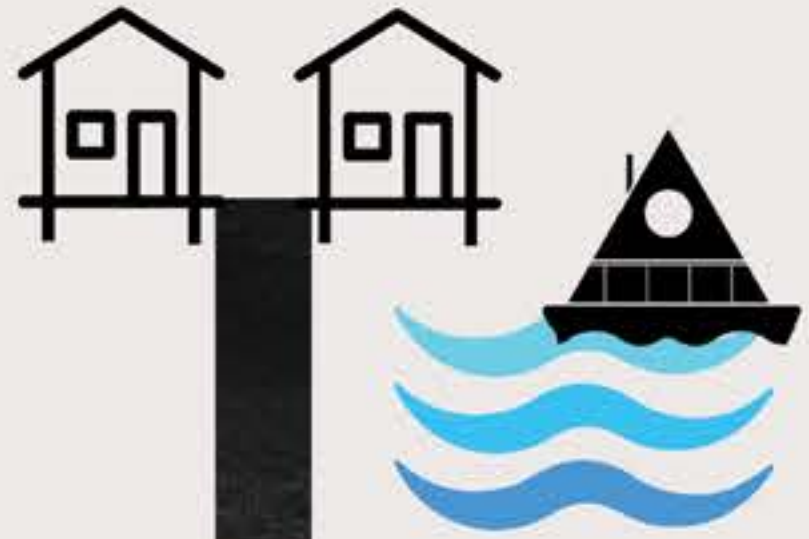
Rewilded Unit

Target: 1 in 10 abandoned homes

Use: Insect wall, planting shed, wildlife blind

Materials: Unpainted timber, moss panels, seed mats

4



A Wall that Moves (Floating and Amphibious Architecture)

20% of new housing will be built on floating or amphibious foundations, integrated into the wall, ensuring flood resilience while maintaining urban density.

5



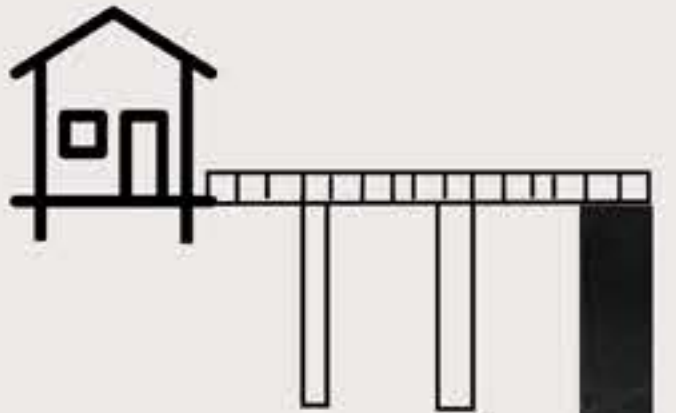
Shell Zone

Applies to: 2m buffer behind tidal wall

Material: Crushed cockle shell substrate

Use: Plant nursery beds, warm seating zone

6



Memory Deck

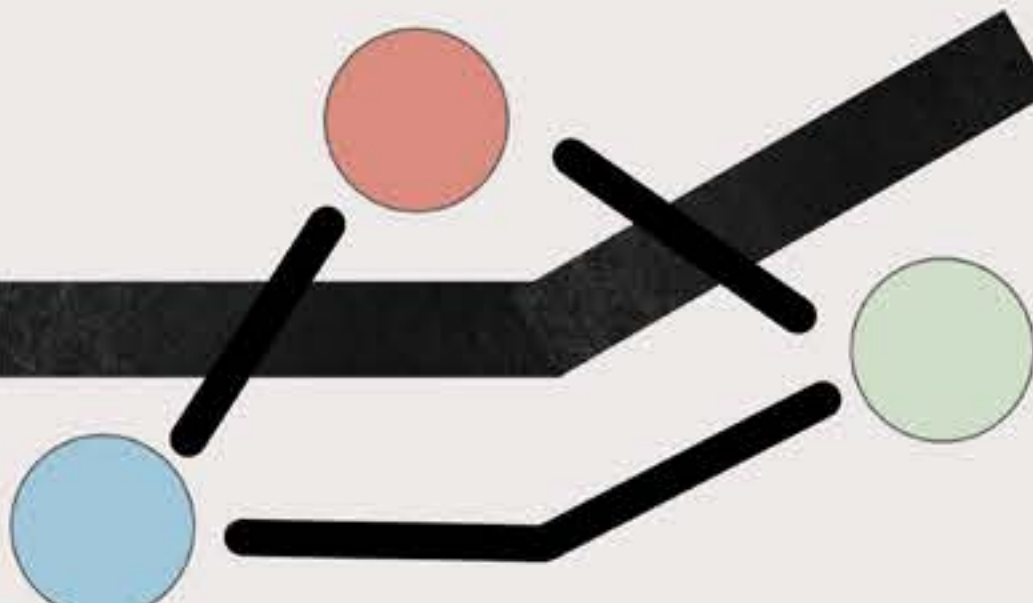
Site: At least 1 platform on re-used timber from prefab porches

Program: Gathering, storytelling, teaching

Placement: Over edge wall with sea views

Labeling: With unit numbers, years of abandonment

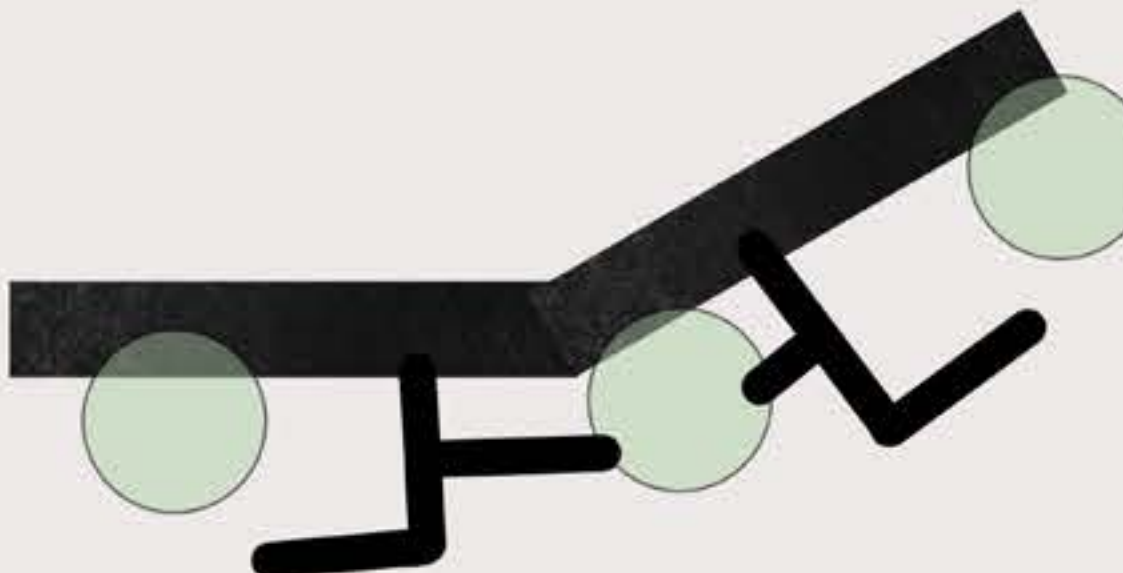
7



A Wall that Connects (New Public and Communal Spaces)

The structure will bridge fragmented communities, integrating pedestrian pathways, elevated walkways, and recreational spaces that ensure the coastline is accessible, not isolated.

8



A Future Expansion Into the Sea (30% Increase in Usable Land)

the wall will expand outward, with floating piers, artificial islands, and modular extensions reclaiming 30% more urban space for housing, biodiversity, and industry. The city does not retreat—it moves forward into the water.

9



Soft Camp

Location: Former static van plots

Program: Tent pads, communal tables, dry toilets

Foundation: Recycled concrete or wood chip beds

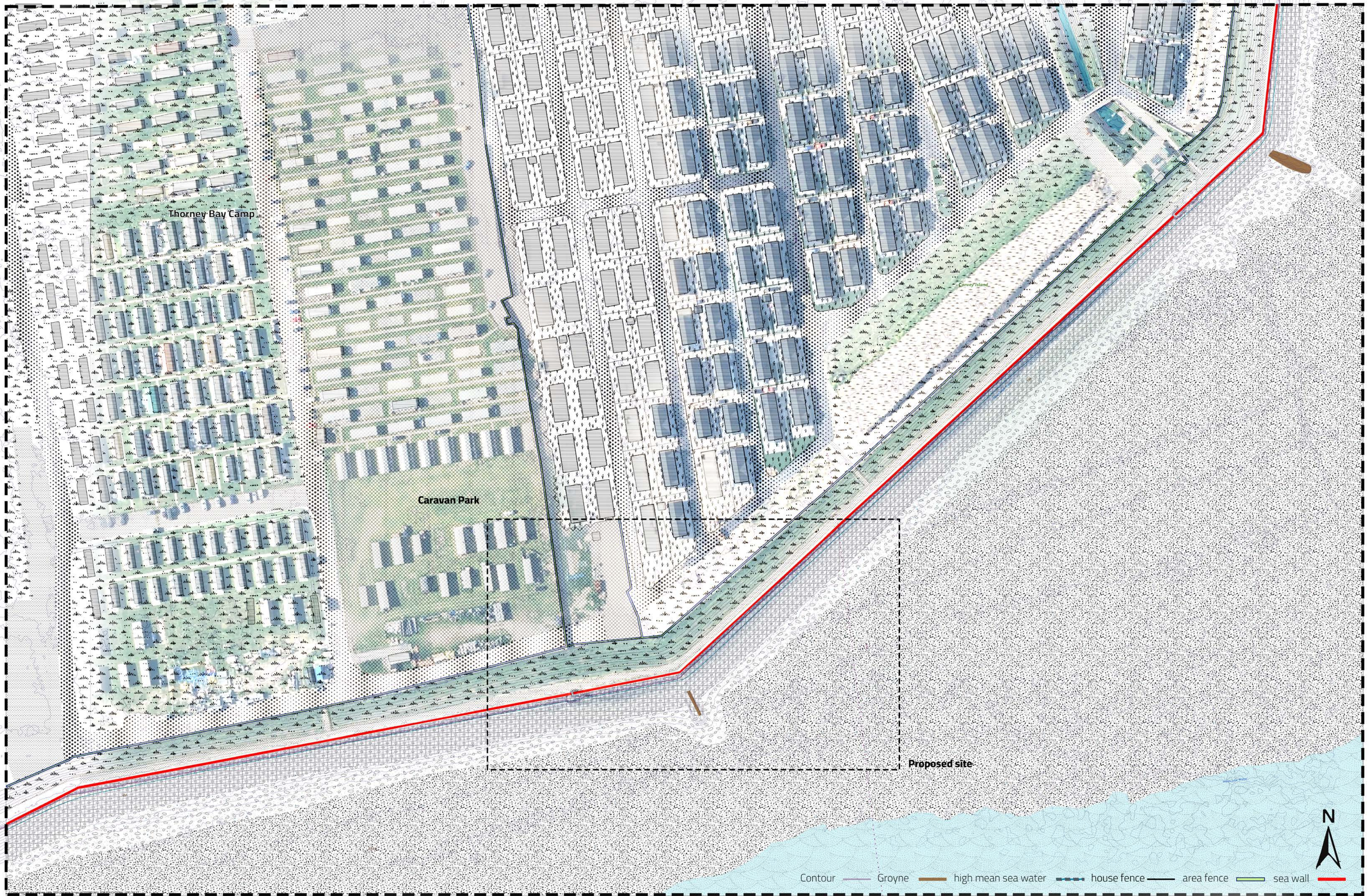
Goal: Low-impact seasonal stay

10



A Wall that Regulates the Sea

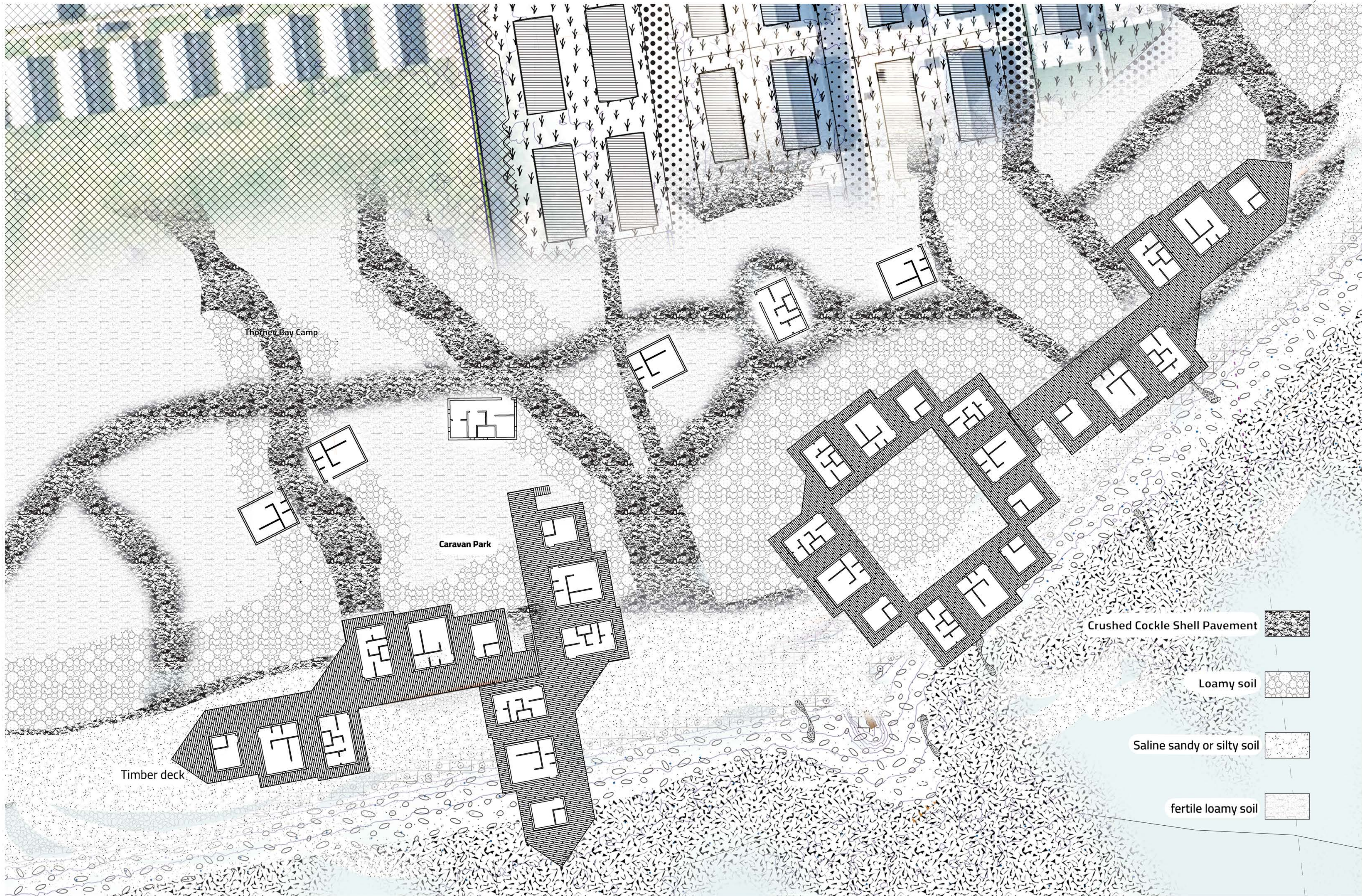
By 2050, soil accumulation will reduce flood intensity by 50% and expand 3km² of intertidal habitat. making the wall a dynamic flood management system.



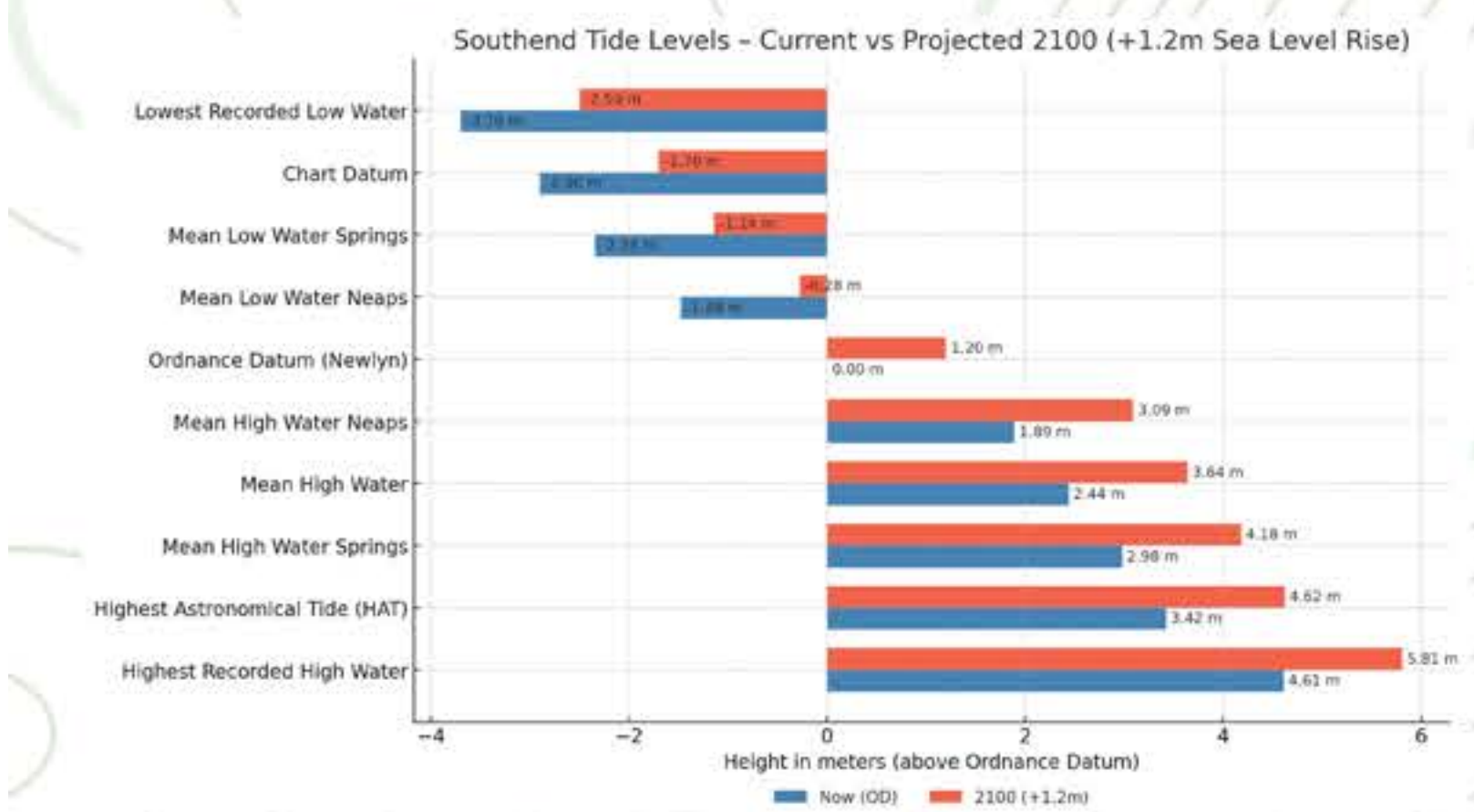
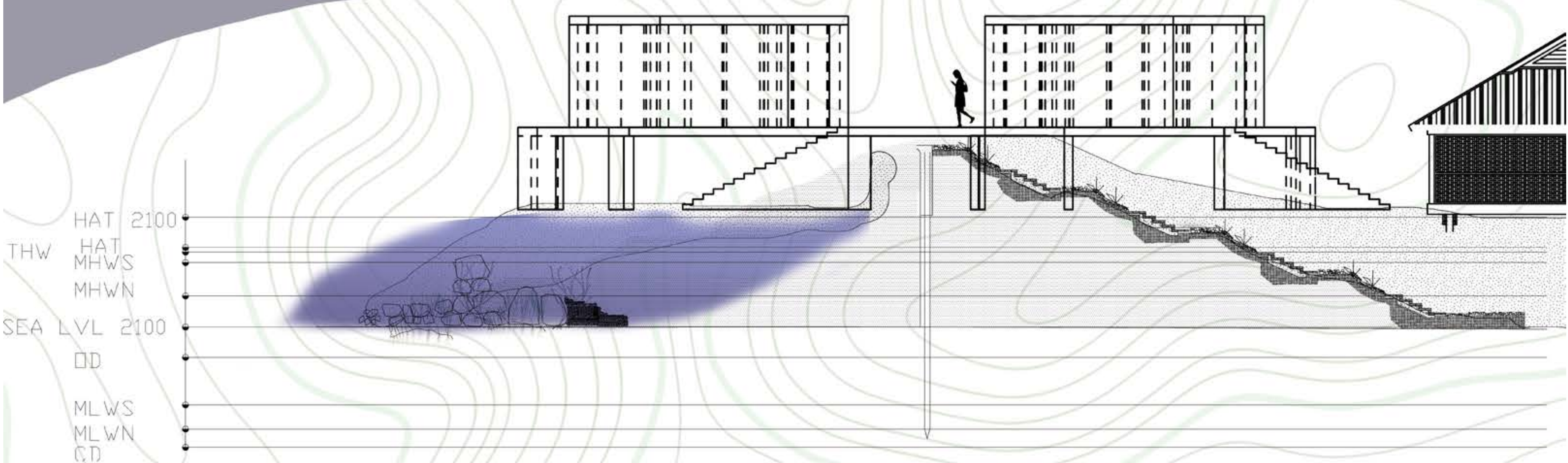
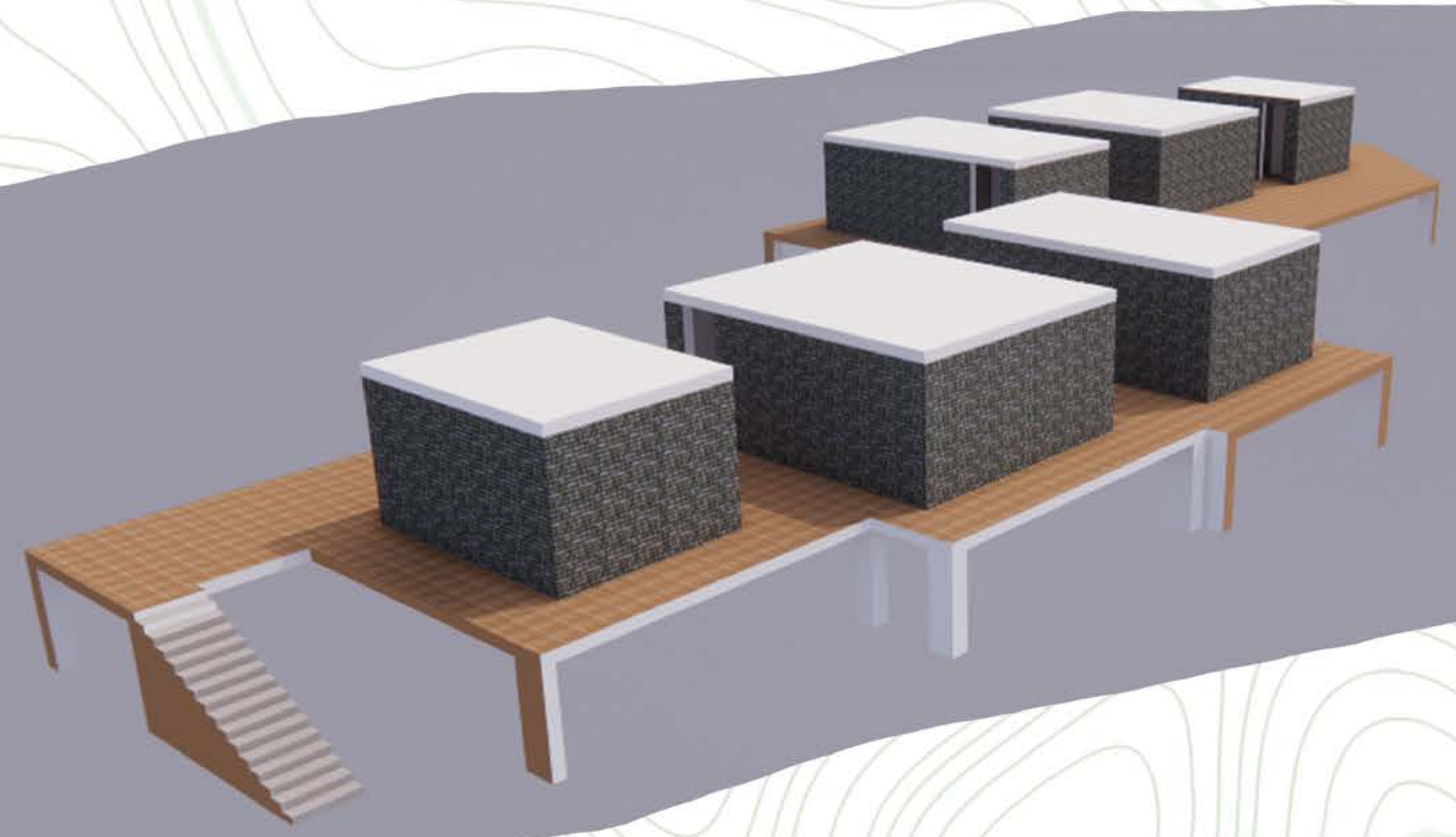
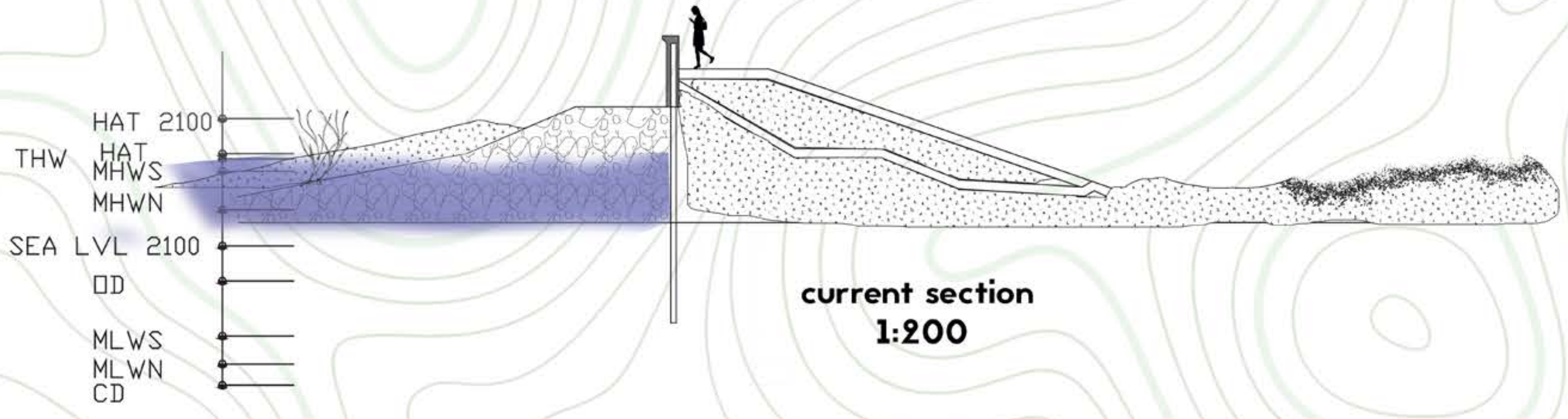
- | | | | | | | | | | | |
|-------------------|--------------------------|-----------|---------------|-----------------|--------------|--------------------|-----|---------|-----------|---------------|
| Construction Zone | Managed Open Green Space | car roads | walking paths | sloping masonry | Inland water | mean low sea water | mud | shingle | Buildings | Private Lawns |
|-------------------|--------------------------|-----------|---------------|-----------------|--------------|--------------------|-----|---------|-----------|---------------|



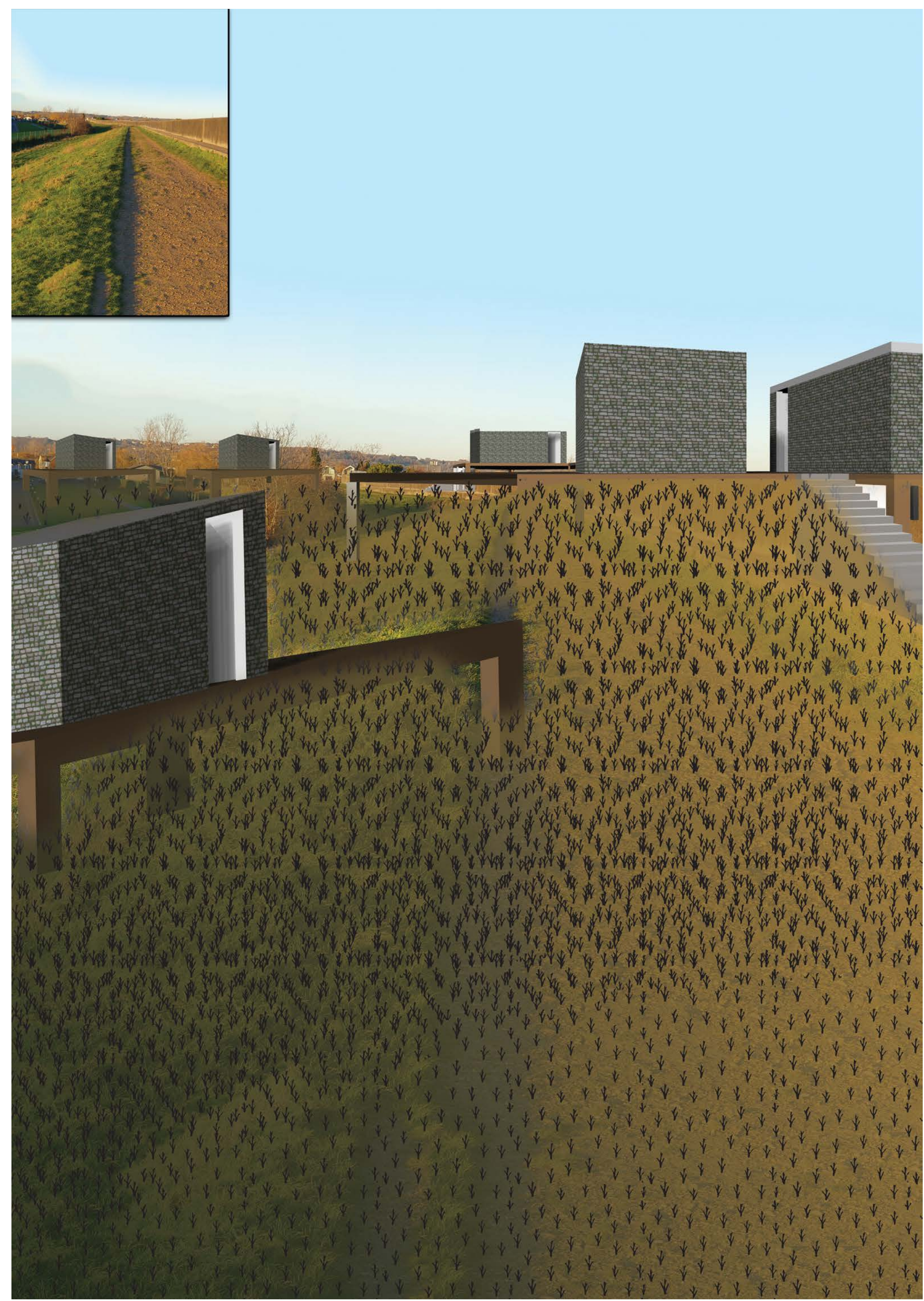


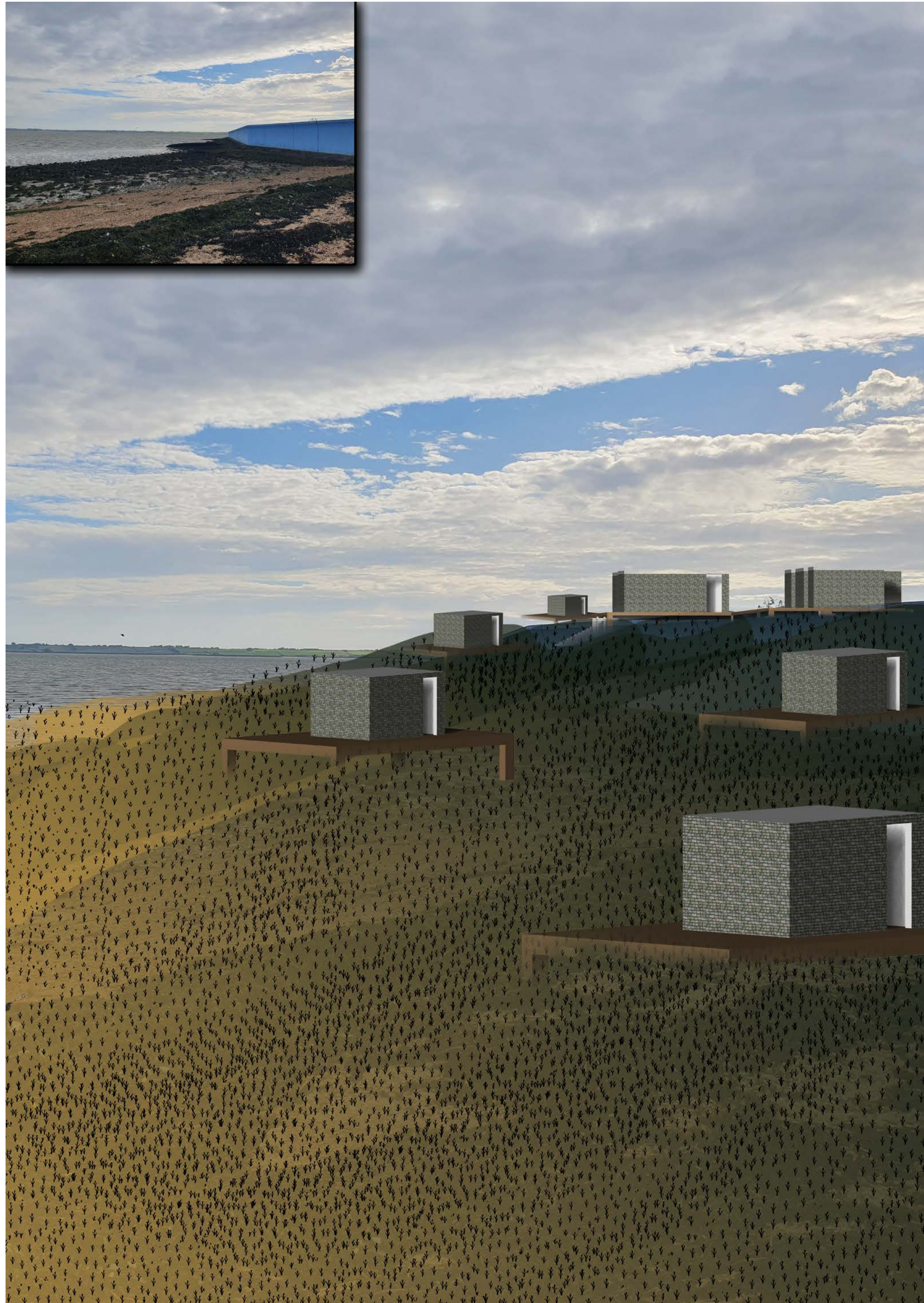
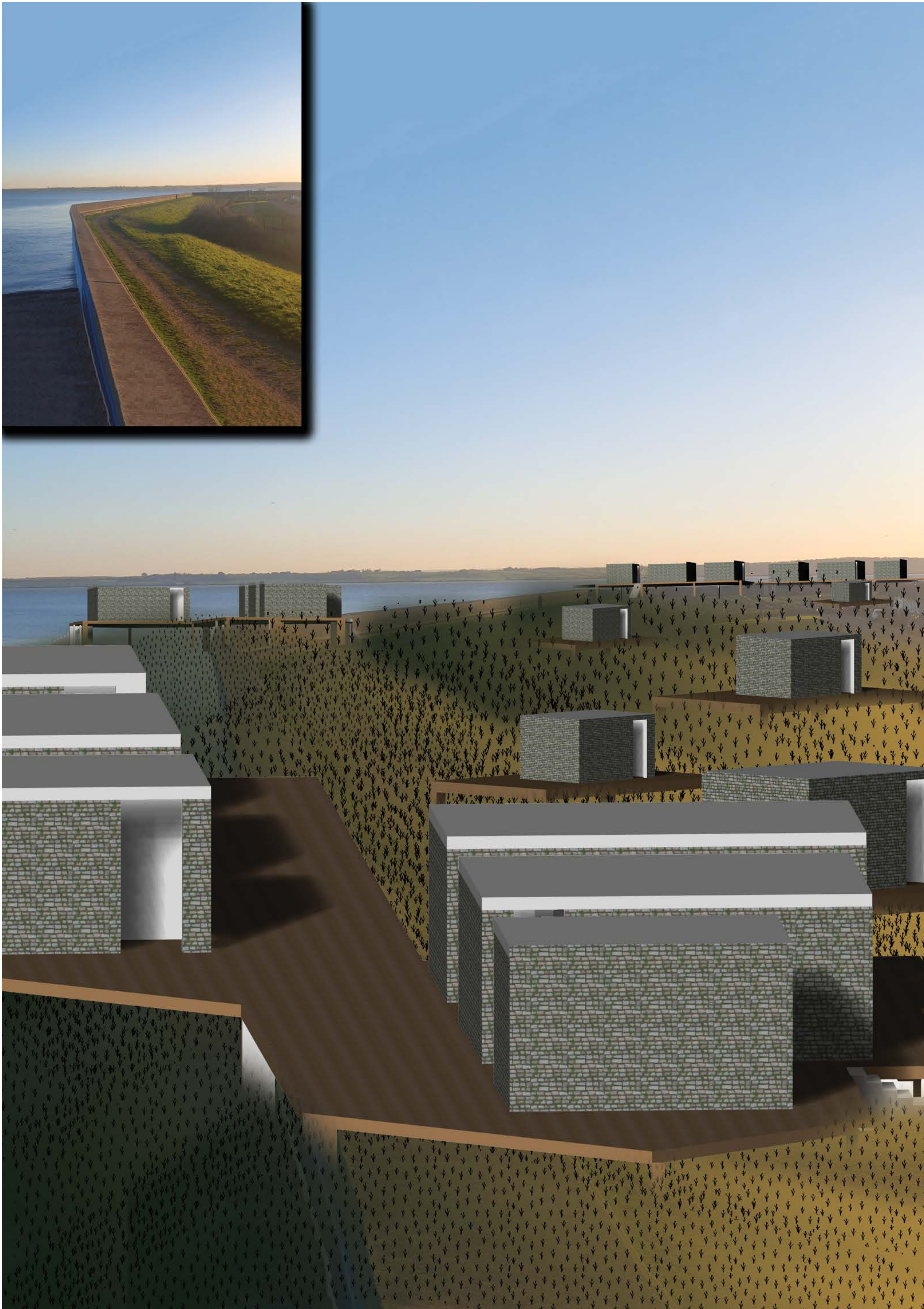


Tide Details, referred to levels at: Southend	Ordance Datum	Chart Datum
Highest Recorded High Water	4.61	7.51 (1953)
Highest Astronomical Tide (HAT)	3.42	6.32
Mean High Water Springs	2.98	5.88
Mean High Water	2.44	5.34
Mean High Water Neaps	1.89	4.79
Ordance Datum (Newlyn)	2.90	2.90
Mean Low Water Neaps	-1.48	1.42
Mean Low Water Springs	-2.34	0.56
Chart Datum	-2.90	-2.90
Lowest Recorded Low Water	-3.70	-0.80 (1920)



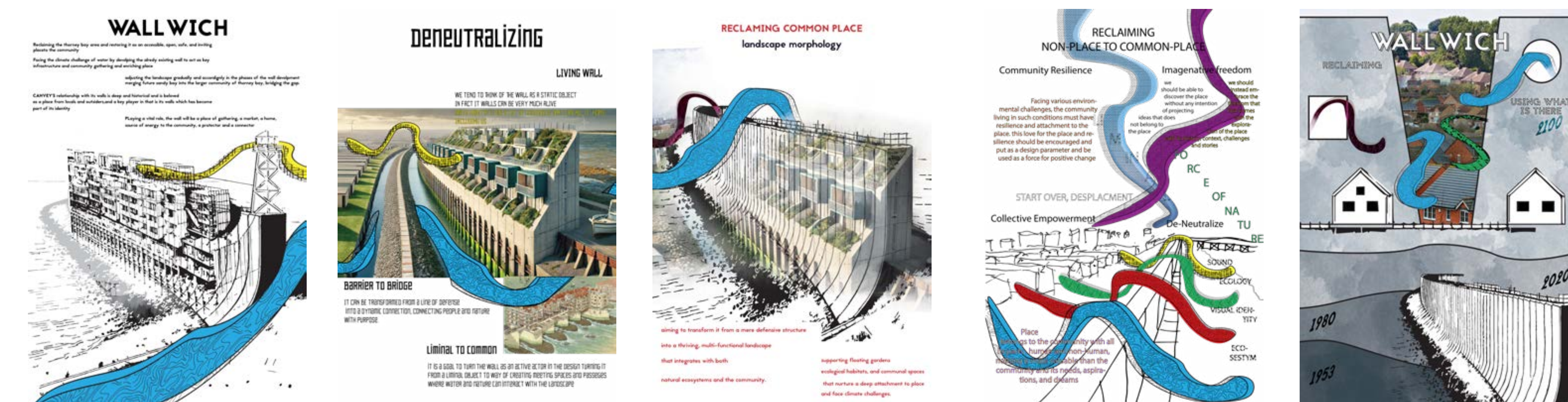
Section drawings
1:200



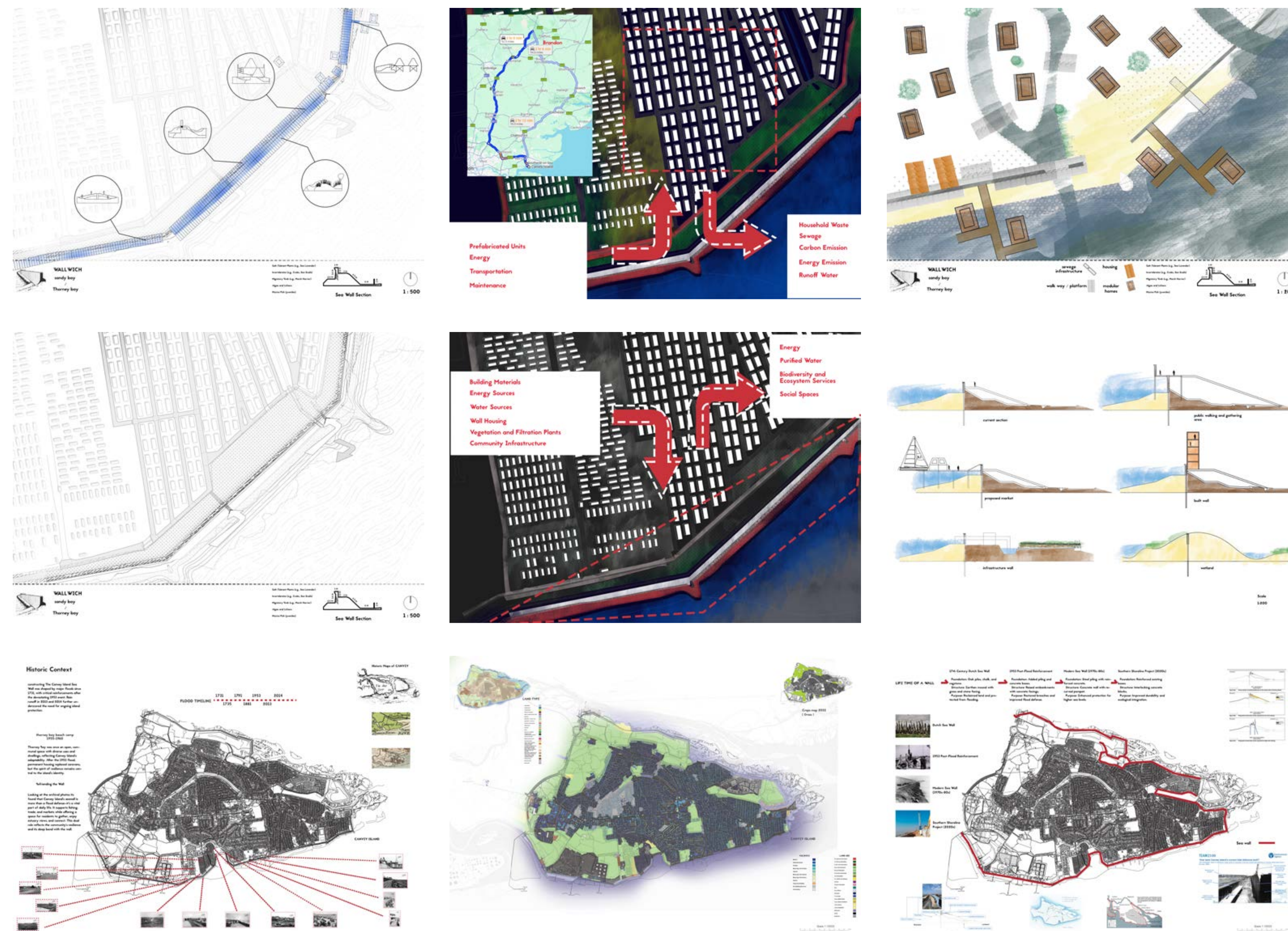


Appendix

Manifestos



Panels



TERM 2- Part 1

Content



TEICHOPOLIS

ENVT-1101-M01-2024-25 Landscape Design 2: Ecologies, Exploration
and Proposition

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Term 2

Manifesto: TEICHOPOLIS

Research: Canvey's wall

Research: It is all inside the wall

Metabolic diagram: The life of a wall

Territorial Plan and section: Canvey as is

Strategic Plan: Canvey commons strategy

Strategic Plan: Canvey commons strategy

Island's values

Research: Wall Community

Research: Site visit analysis

Research: Site visit analysis

Metabolic elements: Rising soil and site activity

Linear Rules

Current site: site and section

Imagined Future: site and section

Building on the past

Safe future

appendix

TEICHOPOLIS

Walls hold history. Walls protect. Walls endure.

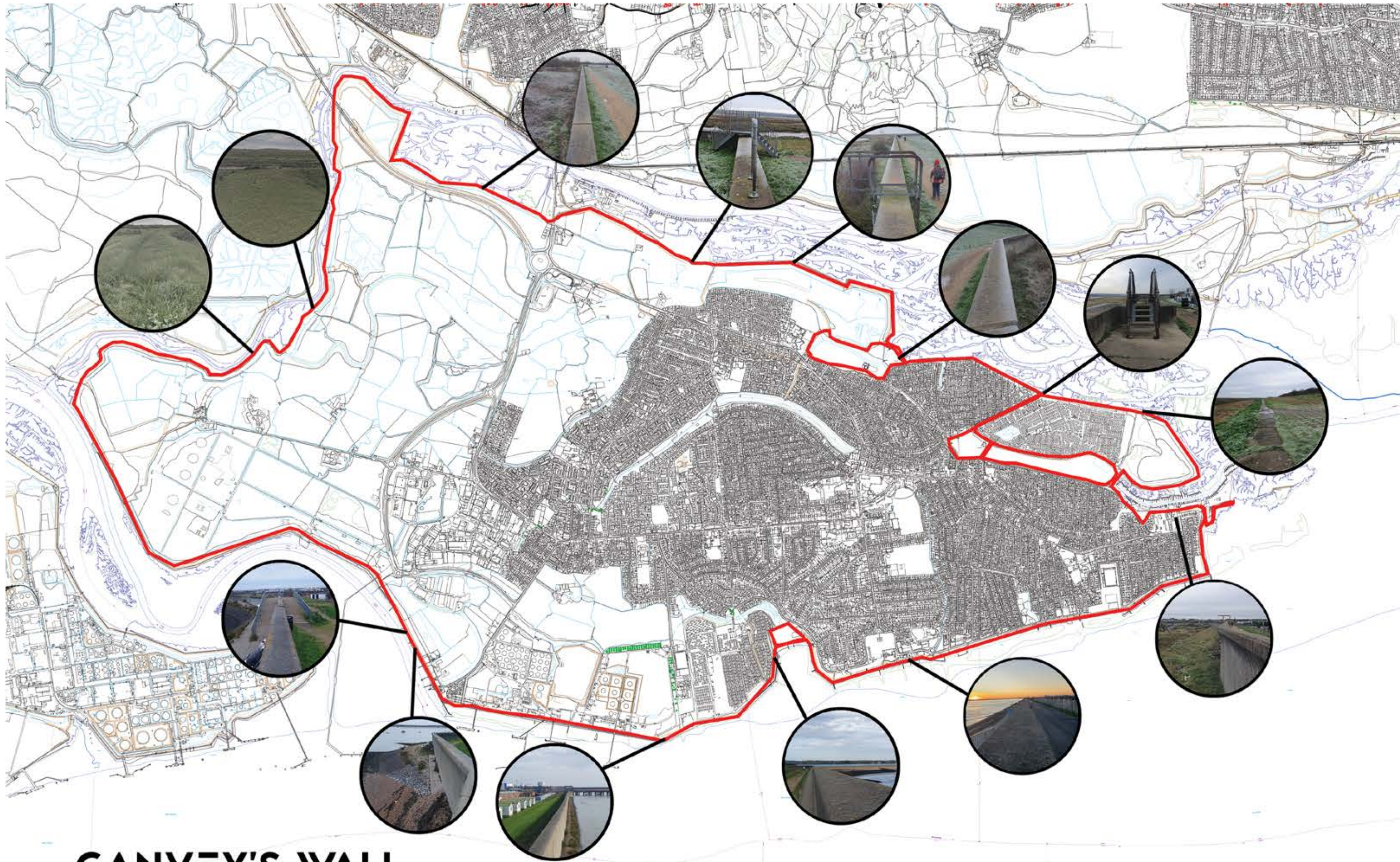
The sea is rising, but Canvey Island will not retreat. Instead of surrendering land to the tides, we **reclaim** it. The seawall is no longer a last line of defense—it is the first **foundation of a new landscape**.

This wall does not just hold back water; it builds land, expands territory, and anchors a new era of urban resilience.

Behind the wall, a living city grows. housing, commerce, and industries designed for stability. On the seaward side, engineered barriers slow waves, capture sediment, and expand Canvey's footprint year by year. The wall is not static; it adapts, rises, and strengthens with time.

If walls must be built, they must do more than defend. They must shape futures, not just protect pasts. This is not an island on borrowed time, it is a city that reclaims its place in the world.

Teichopolis is not a retreat. It is a declaration: **the land remains**.



CANVEY'S WALL

- LIFE TIME OF A WALL**
- 17th Century Dutch Sea Wall**
Foundation: Oak piles, chalk, and ragstone.
Structure: Earthen mound with grass and stone facing.
Purpose: Reclaimed land and protected from flooding.
 - 1953 Post-Flood Reinforcement**
Foundation: Added piling and concrete bases.
Structure: Raised embankments with concrete facings.
Purpose: Restored breaches and improved flood defense.
 - Modern Sea Wall (1970s-80s)**
Foundation: Steel piling with reinforced concrete.
Structure: Concrete wall with recurved parapet.
Purpose: Enhanced protection for higher sea levels.
 - Southern Shoreline Project (2020s)**
Foundation: Reinforced existing bases.
Structure: Interlocking concrete blocks.
Purpose: Improved durability and ecological integration.



Dutch Sea Wall



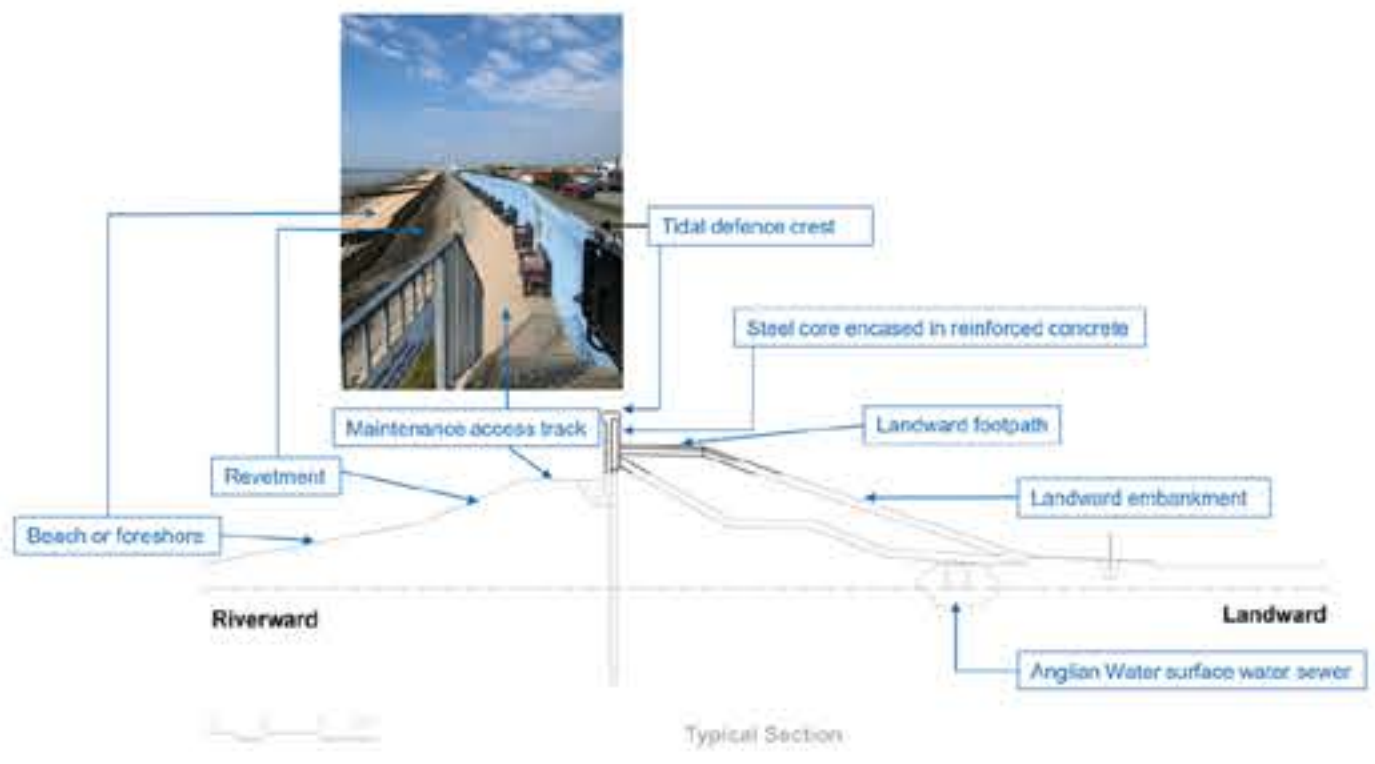
1953 Post-Flood Reinforcement



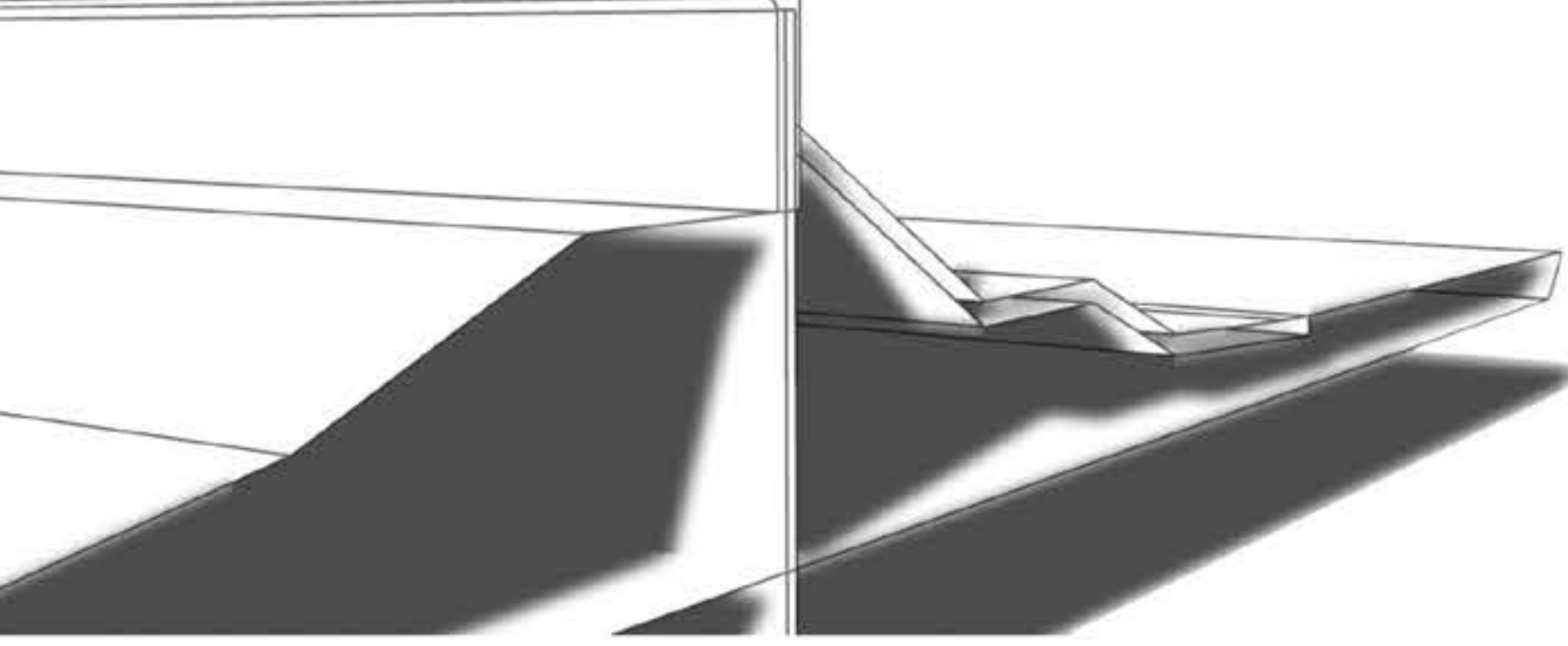
Modern Sea Wall (1970s-80s)



Southern Shoreline Project (2020s)



Typical Section



TEAM2100

How were Canvey Island's current tidal defences built?

This photograph, taken in Hothaven Creek, gives an example of how the current tidal defences on Canvey Island were built in the early 1980s.

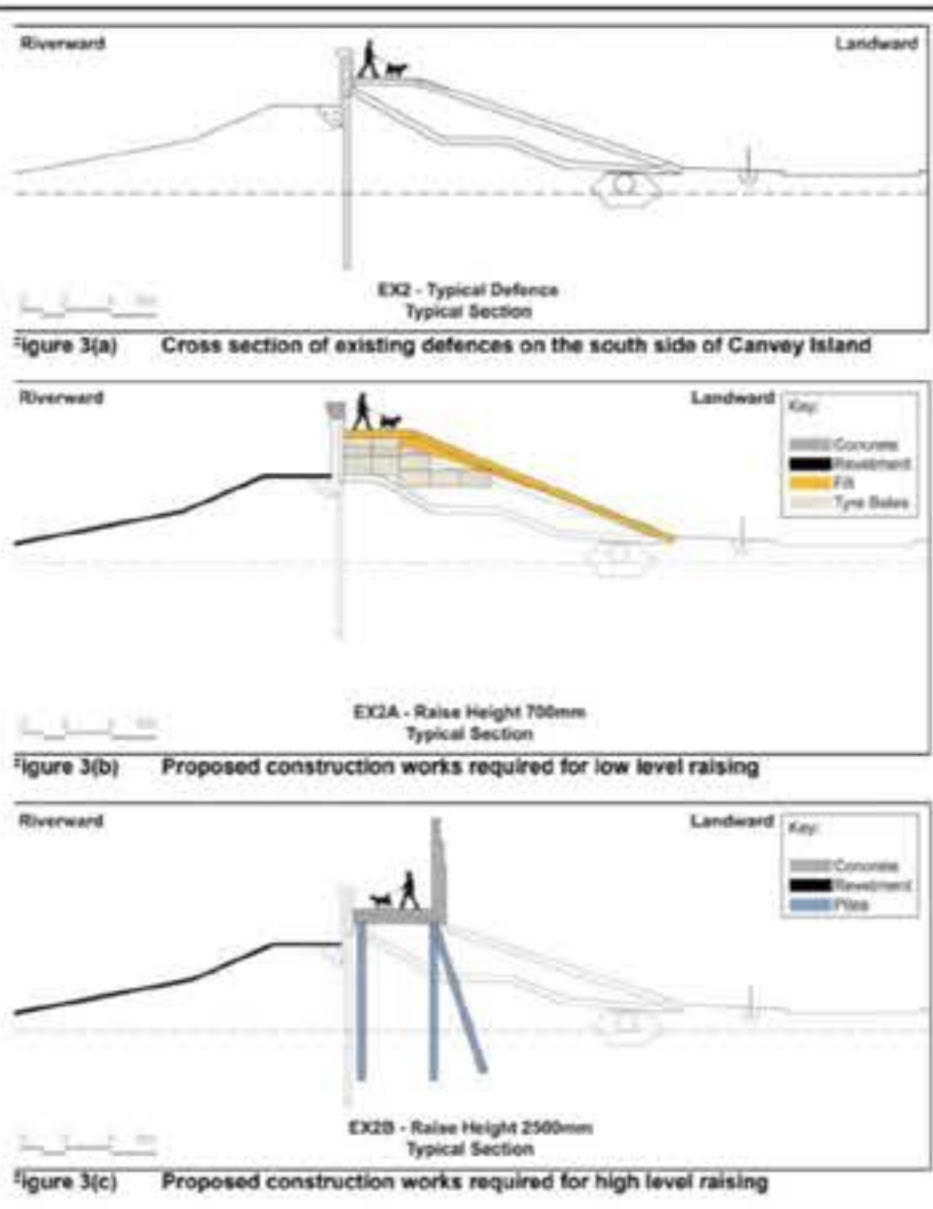


Figure 3(a) Cross section of existing defences on the south side of Canvey Island

Figure 3(b) Proposed construction works required for low level raising

Figure 3(c) Proposed construction works required for high level raising

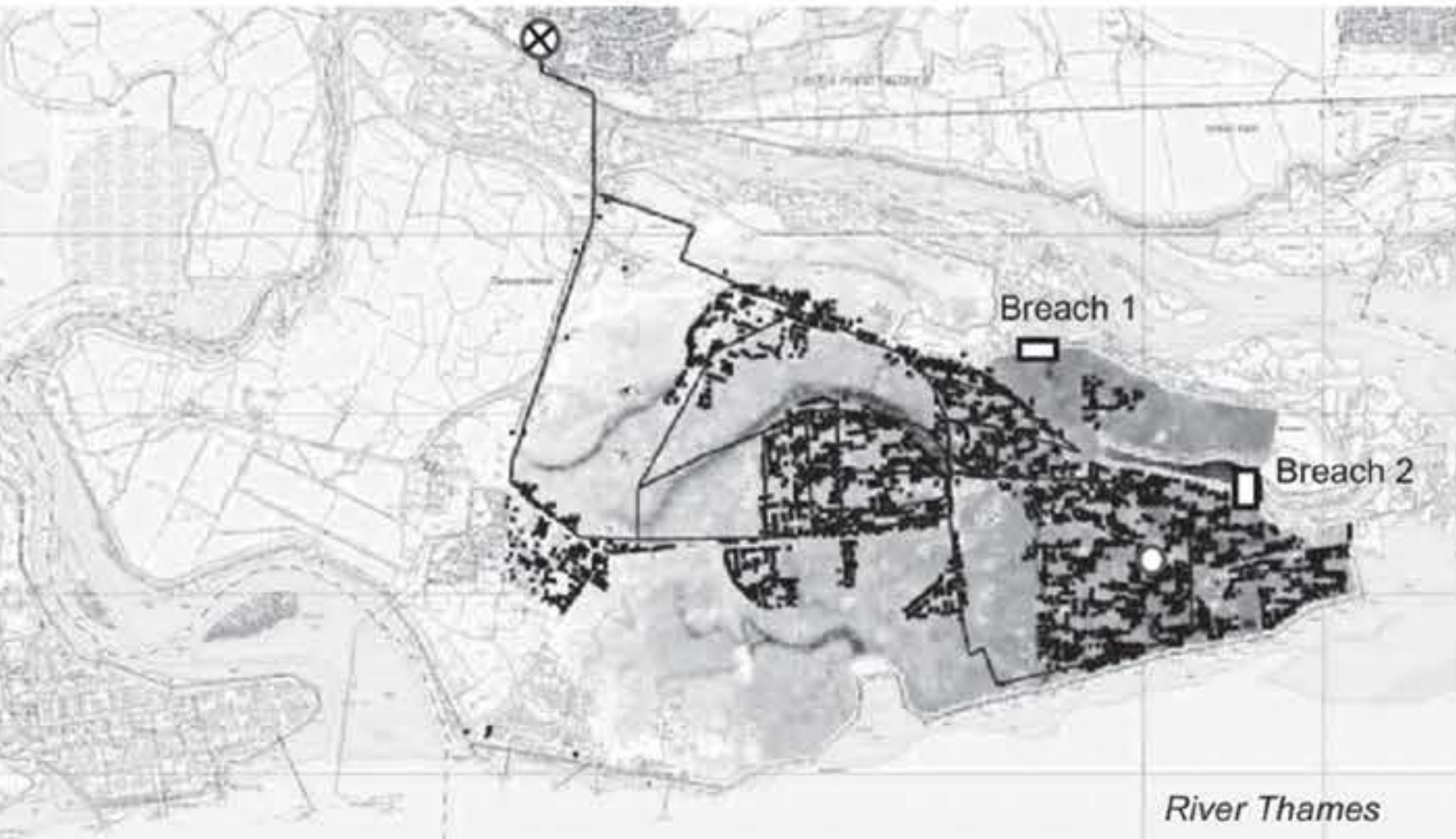
Feature	Spec
Length	~23 km (14 miles) — wraps around most of the island
Current Height	~5.5 metres above sea level (AOD)
Crest Width (top)	~2.5 to 3.0 metres
Base Width (bottom)	~4 to 6 metres (varies depending on section)
Material	Reinforced concrete + stone revetment (old Essex blocks)
Updated Material	Open Stone Asphalt (OSA) — more durable, erosion-resistant
Slope	1:2 to 1:3 (angled sides for stability in some sections)

IT IS ALL INSIDE THE WALL

Canvey Island has always been defined by water—and by the wall that holds it back. Once a scattering of tidal islets, the land was consolidated through 17th-century Dutch engineering. Today, a concrete sea wall encircles the island, physically separating life from the Thames. Historical maps show what was gained; flood simulations reveal what could still be lost.

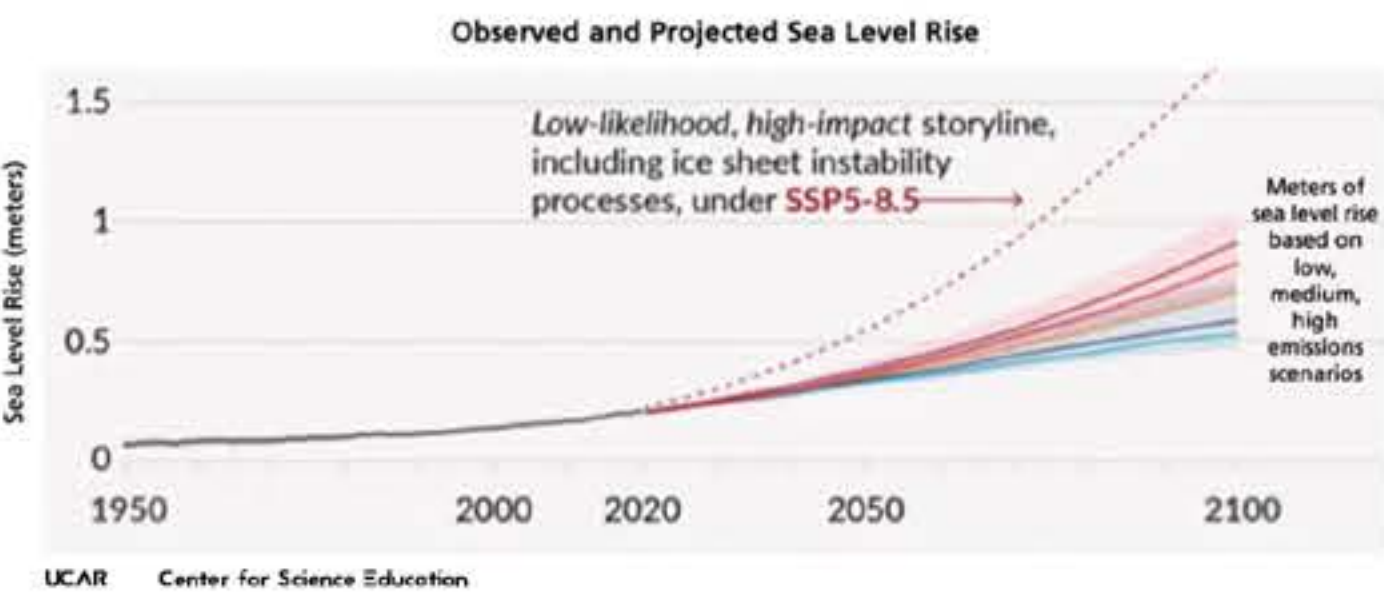
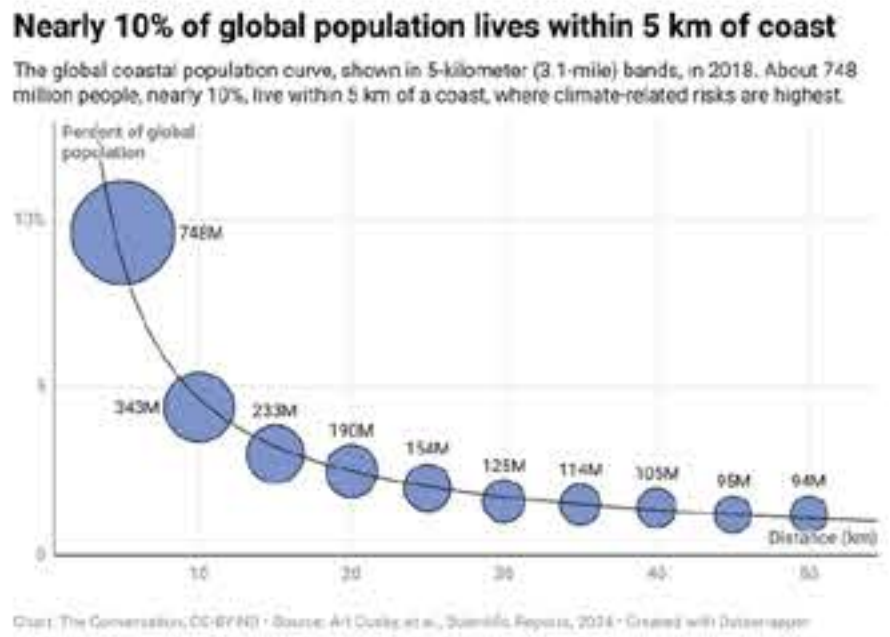
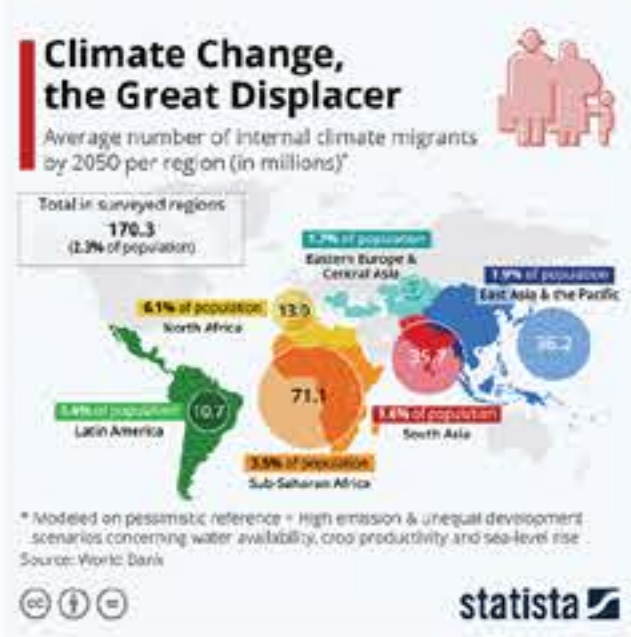


2025 map

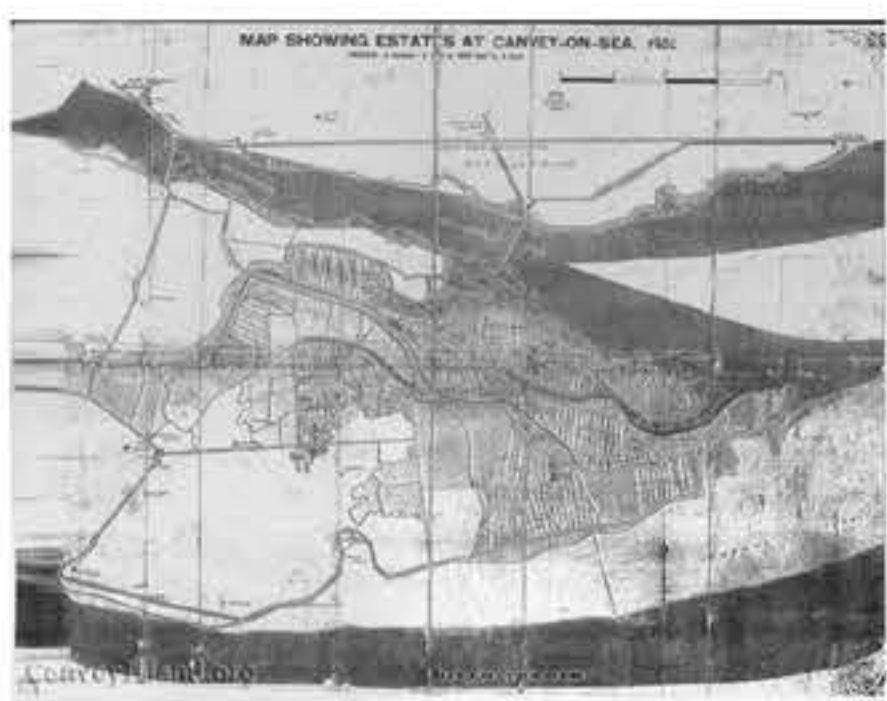


Breach points of the 1953 flood

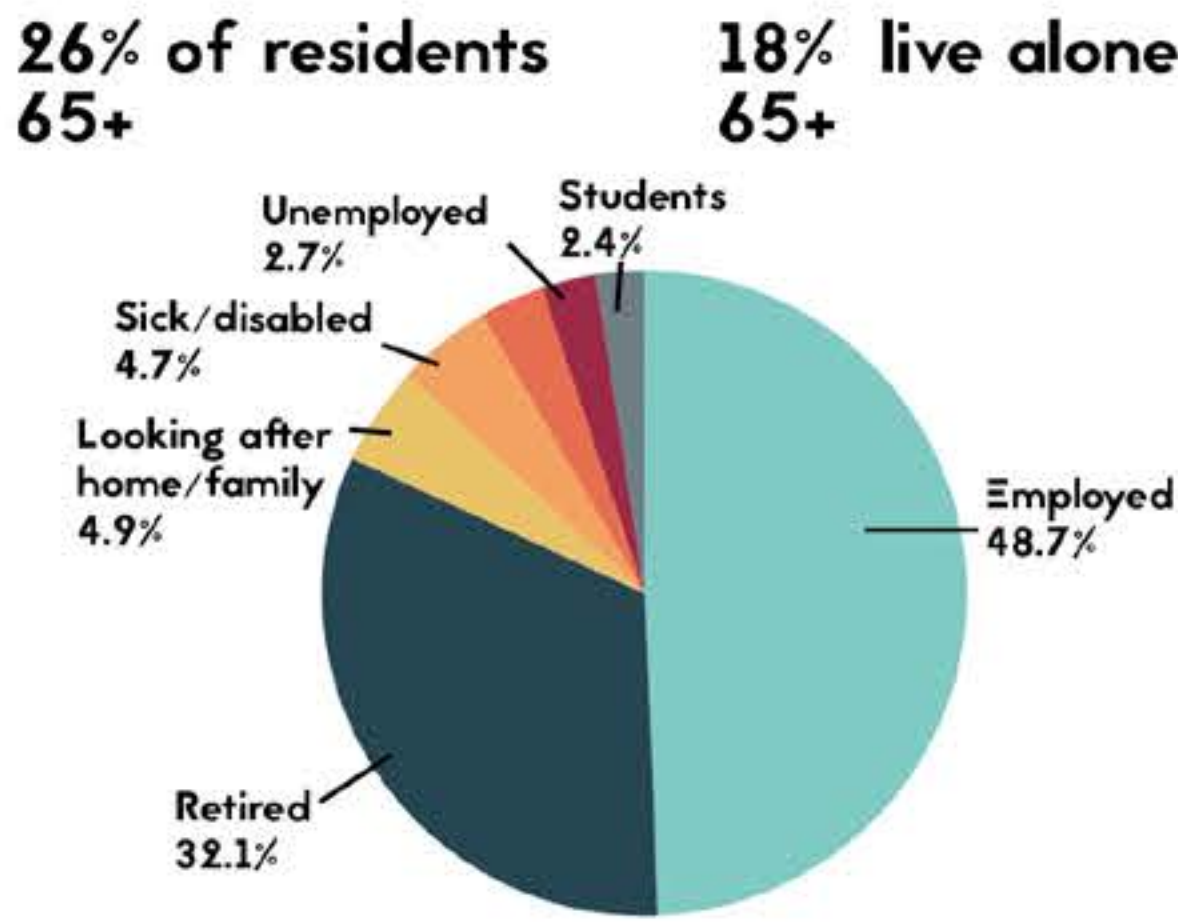
But as sea levels rise and housing crises deepen in urban cores like London, Canvey's vulnerability coexists with potential. With 74–82% home ownership and spacious plots, the island may become a site of retreat or resettlement.



National projections suggest 1 in 4 people will be over 65 in two decades, urban populations will grow from 54% to 68% by 2050, and deaths will increase from 2.6m to 4m annually.



Now home to 38,003 residents , Canvey functions as a low-rise retirement suburb.



93% live in detached, semi-detached, or terraced homes

79% of households own a car

traits of a quiet, car-dependent landscape.

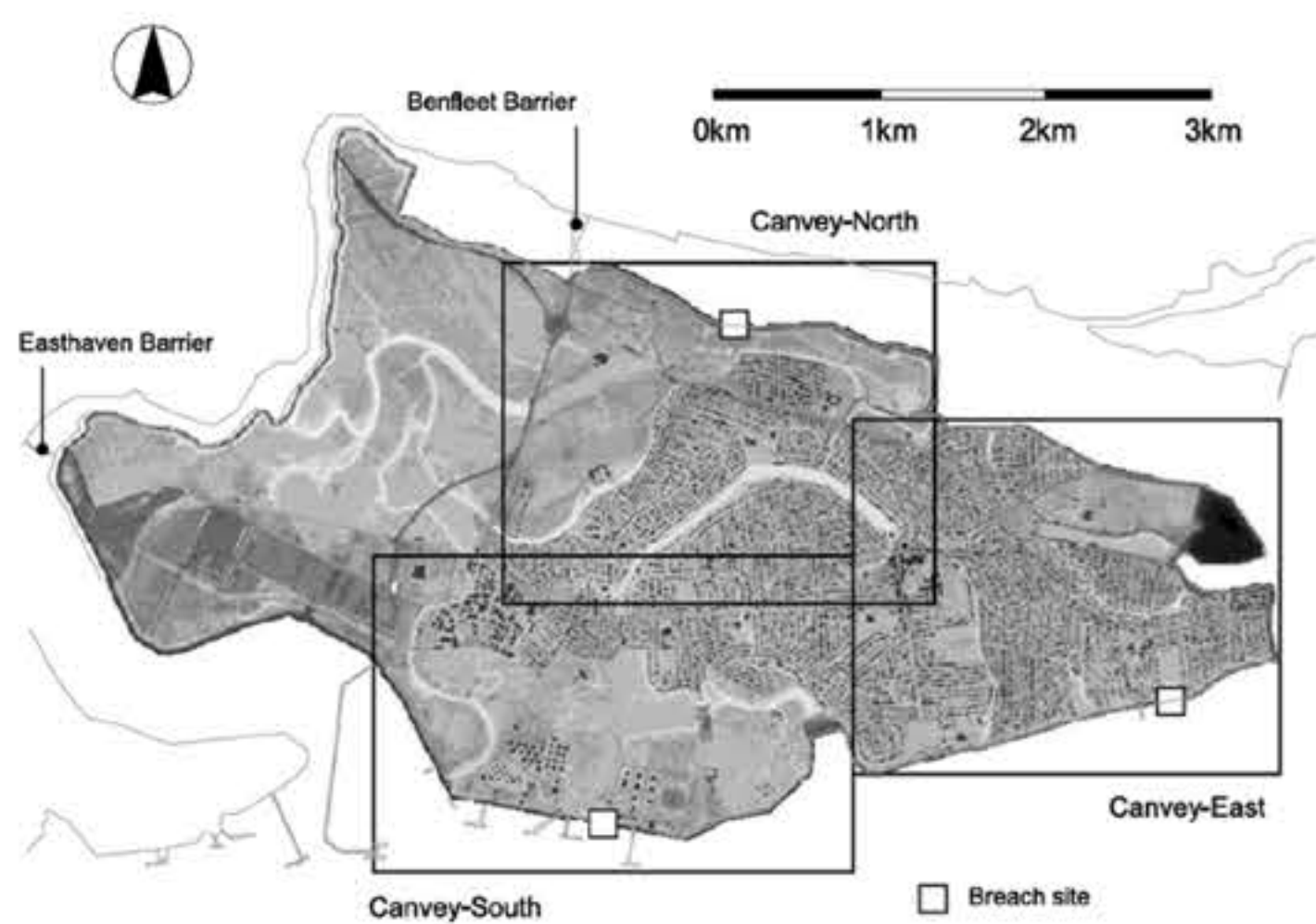


Figure adapted from: Bates, P. D., Horritt, M. S., & Fawcett, T. J. (2010). Water Resources Research, 46(6). https://doi.org/10.1029/2005WR004597 © American Geophysical Union.

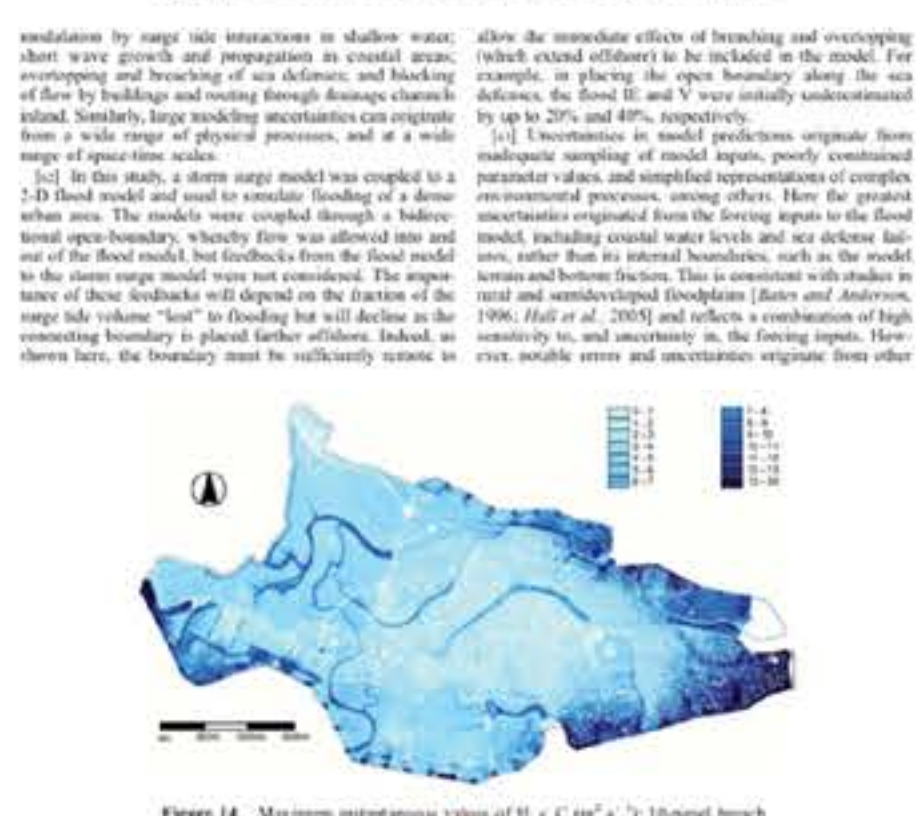
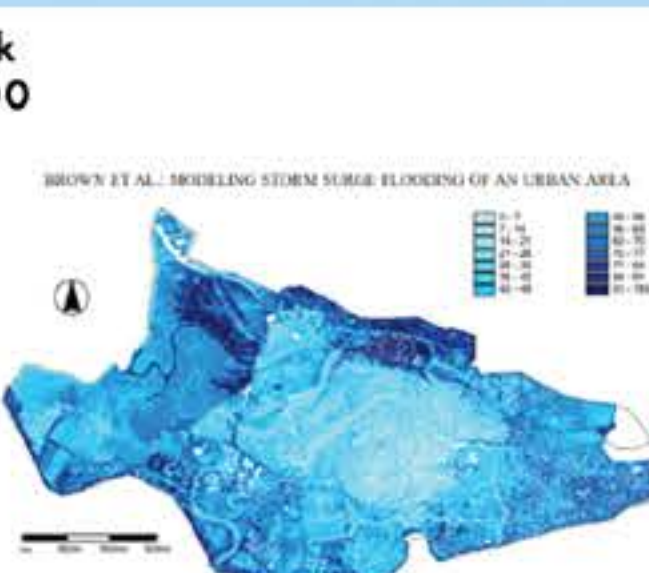
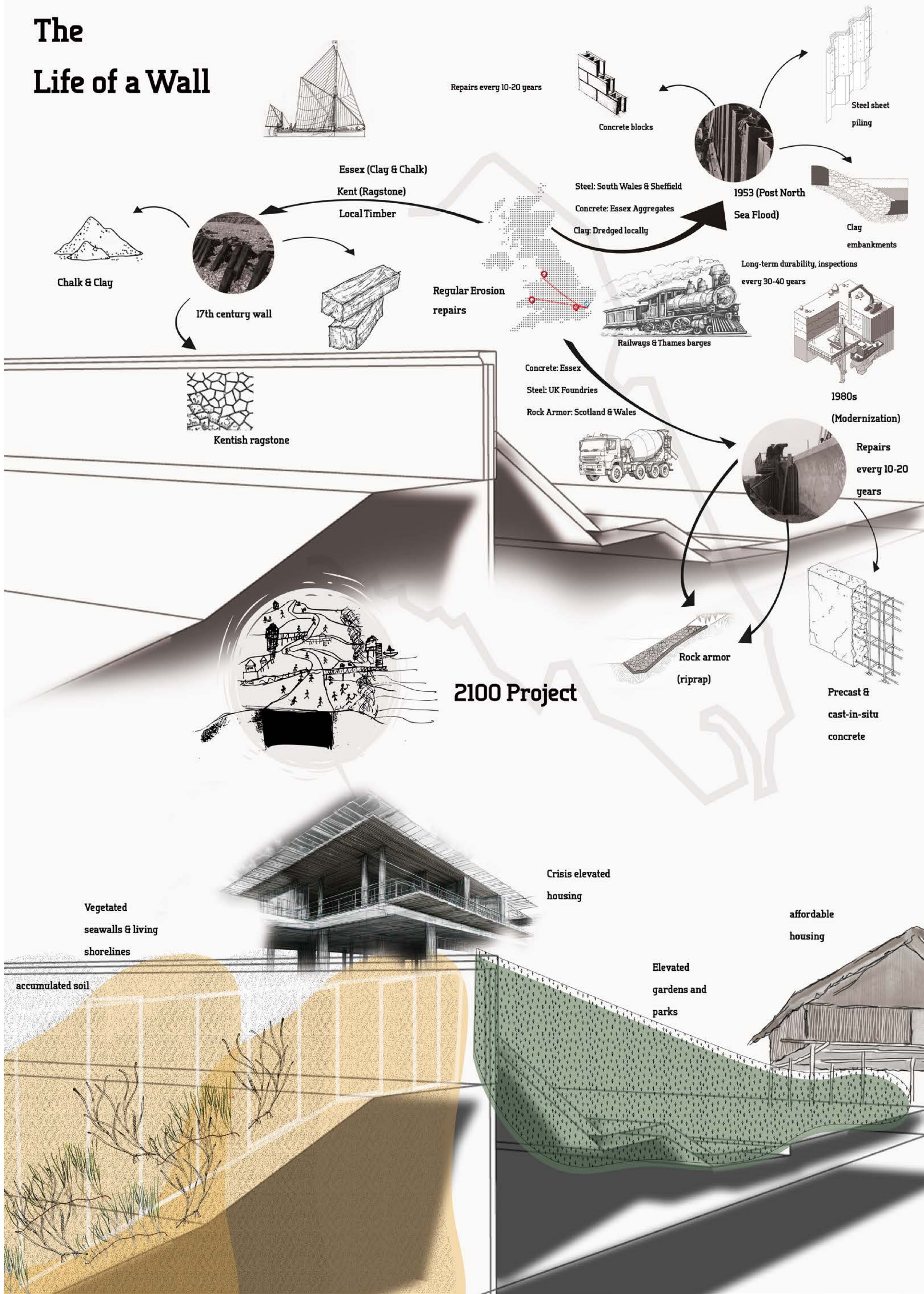
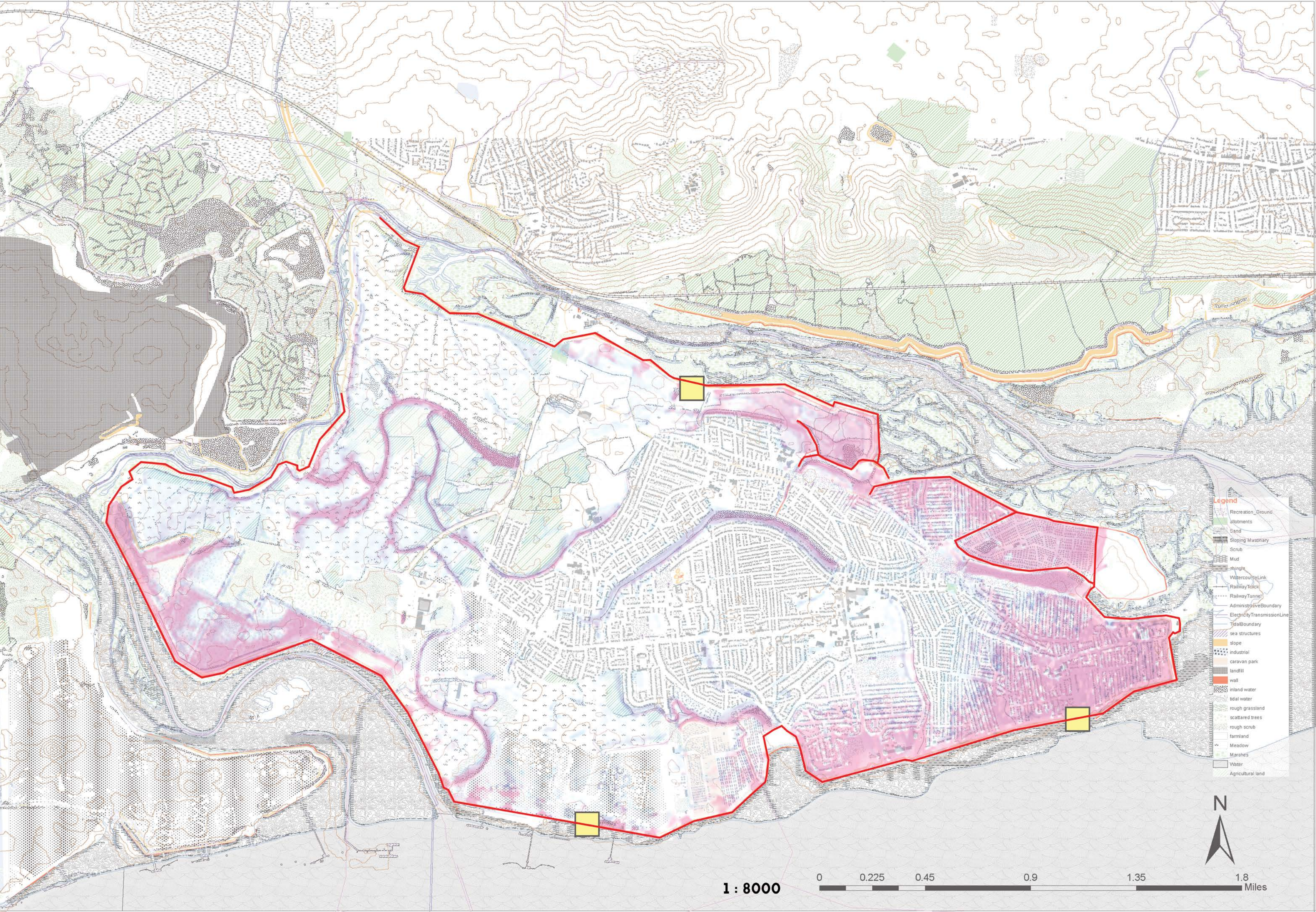


Figure adapted from: Bates, P. D., Horritt, M. S., & Fawcett, T. J. (2010). Water Resources Research, 46(6). https://doi.org/10.1029/2005WR004597 © American Geophysical Union.

The Life of a Wall



Canvey as is



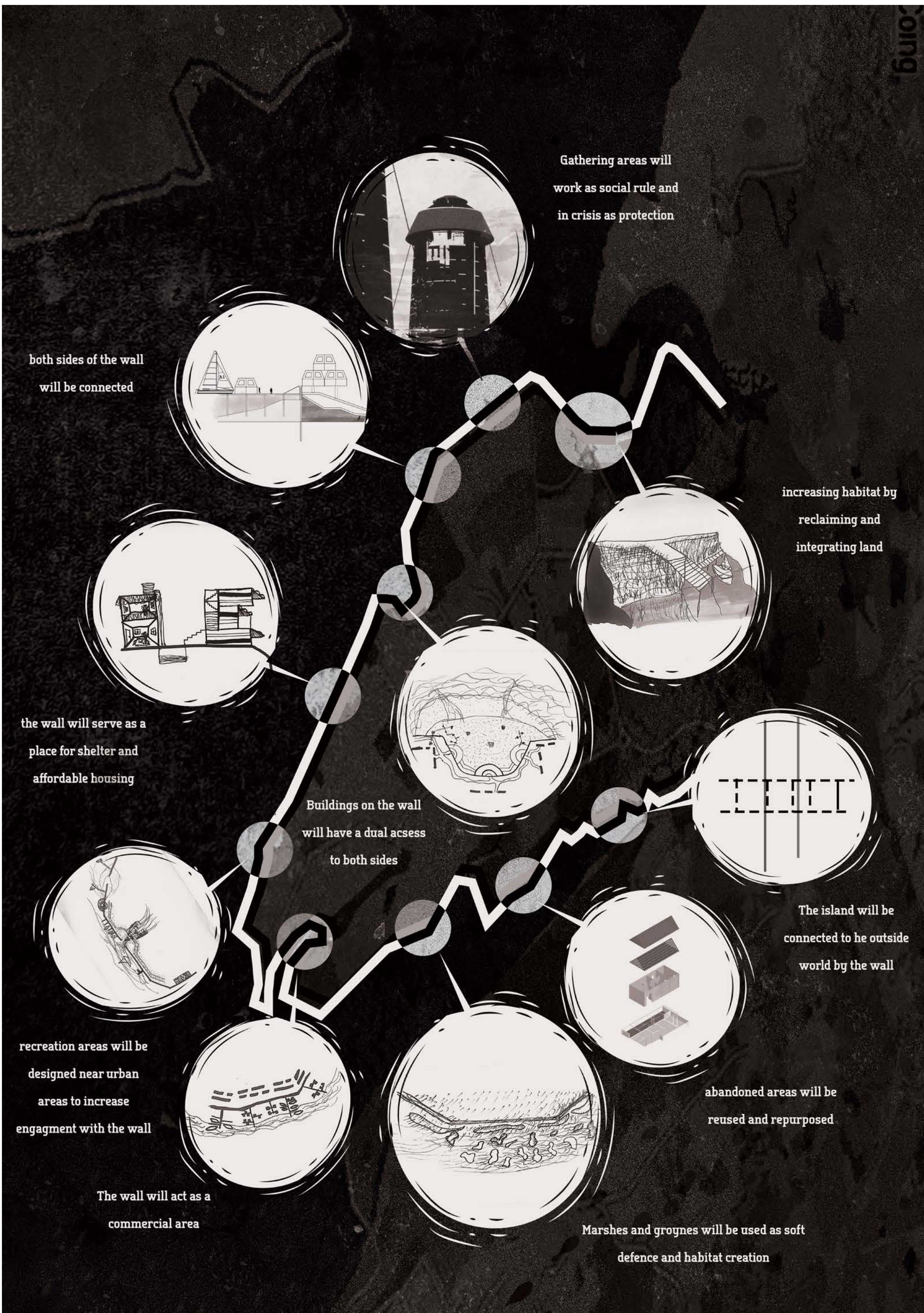
Breach
Point
Pressure
area
Water
depth
and
speed

1 : 500

CANVEY COMMONS STRATEGY

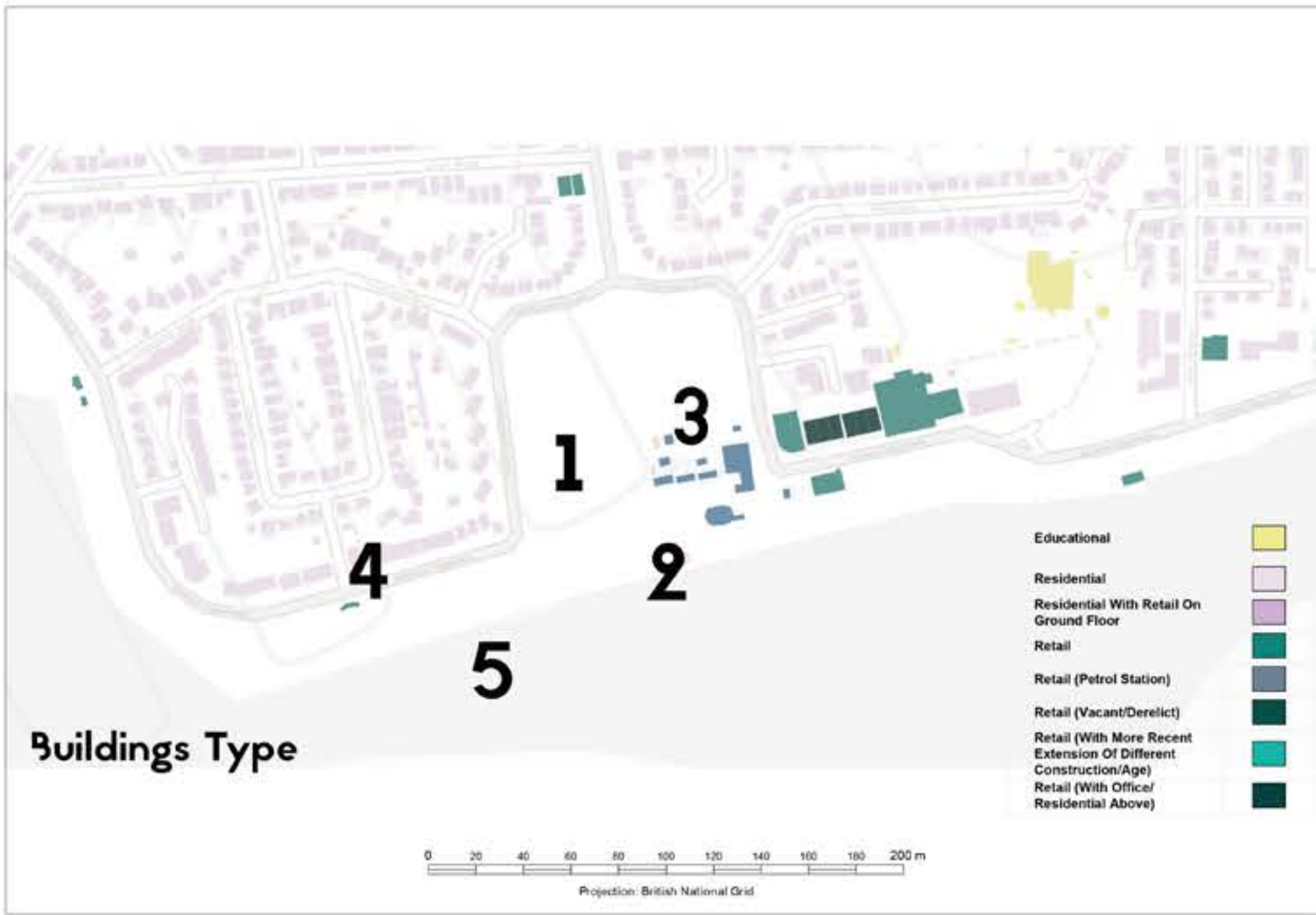


1: 8000



TERM 2- Part 2

Wall Community



Contextual Overview:

- Location: Canvey Island Seafront, encompassing Western/Eastern Esplanade and Labworth Park.
- Key Features:
- 1- Canvey Central Park & Memorial Garden: Spaces of reflection and community gathering.
 - 2- Labworth Café: A 1930s Modernist icon designed by Ove Arup, representing the fusion of engineering and leisure.
 - 3- Fantasy Island Amusement Park: A symbol of recreational activity juxtaposed against the backdrop of flood defenses.
 - 4- Residential Edges: Homes backing onto the civic coastline, highlighting the intersection of private and public realms.
 - 5- Sea Wall & Linear Beach: Physical manifestations of flood defense infrastructure.

Historical Context:

The Labworth Café, constructed in 1932–1933, is the only building solely designed by engineer Ove Arup. Its design resembles the bridge of the Queen Mary ship, symbolizing maritime heritage and modernist architecture.

The North Sea Flood of 1953 devastated Canvey Island, leading to the construction of extensive sea defenses. The current sea wall, completed in the 1980s, is a response to this historical event.



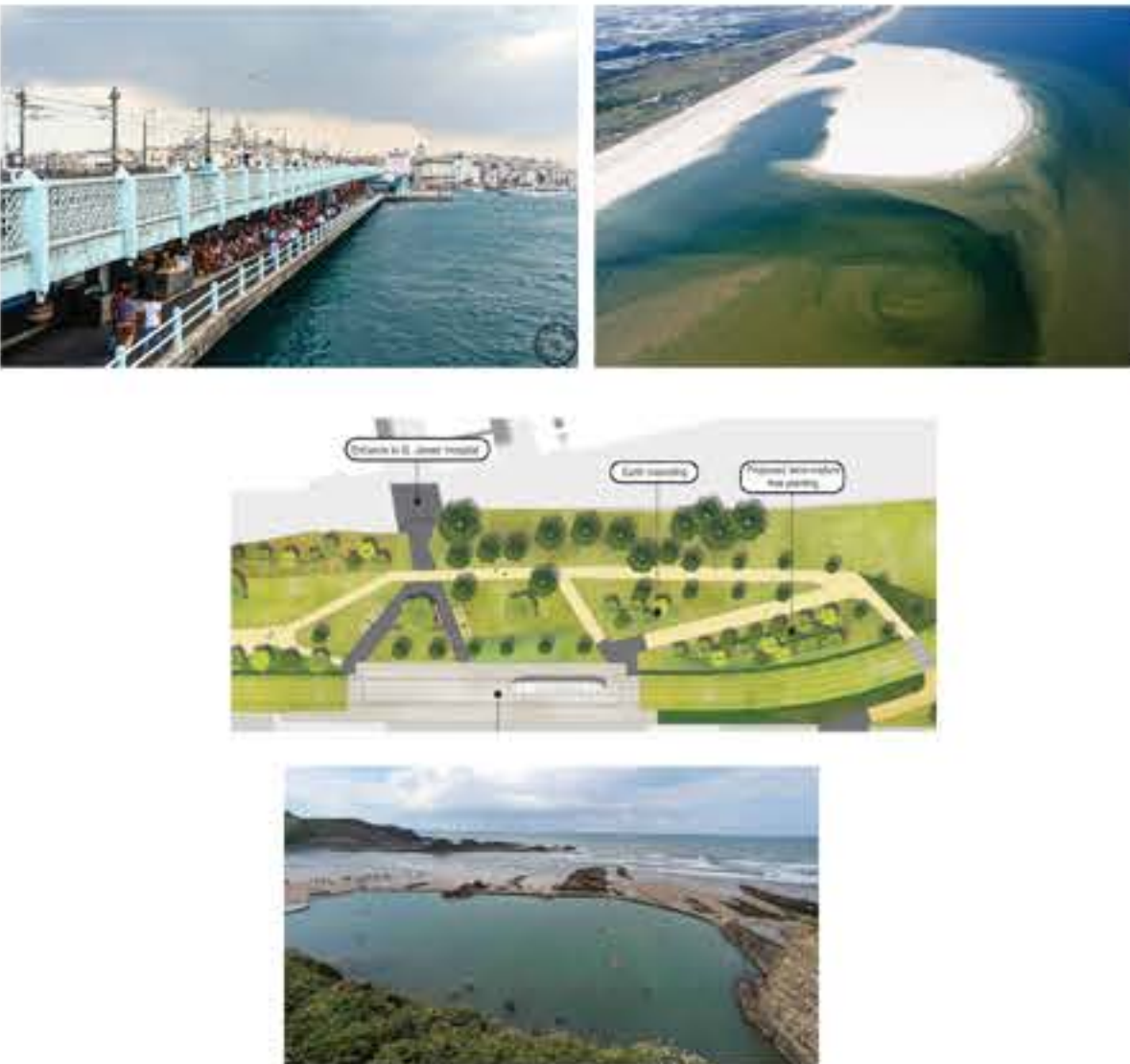
Design Concept:

In anticipation of rising sea levels, this proposal creates an artificial civic hill along Canvey Island's seafront. The existing sea wall is embedded and extended using reclaimed soil and soft infrastructure. This new terrain supports public life, biodiversity, and adaptive housing, creating a hybrid condition between protection and participation.

The iconic Labworth Café, instead of being isolated or buried, becomes the centerpiece of a new landform—partly surrounded by the sloping hill, with a dual entrance: one from the original promenade and a new one from the elevated ridge. The café's program is extended to accommodate public events, education, and a terrace garden. A manmade beach and wetland belt buffer the tidal edge.

Key Moves:

- Build up soil over time from demolition debris and dredge surplus.
- Remove the car road, replacing it with elevated pedestrian/bike paths.
- Embed housing, kiosks, and gathering spaces into the slope.
- Connect the café spatially with Canvey Central Park and the promenade.
- Use native planting, dunes, and sand-retaining groynes to soften the coast.



SITE VISIT ANALYSIS

Tidal Edge / wet-dry zone:



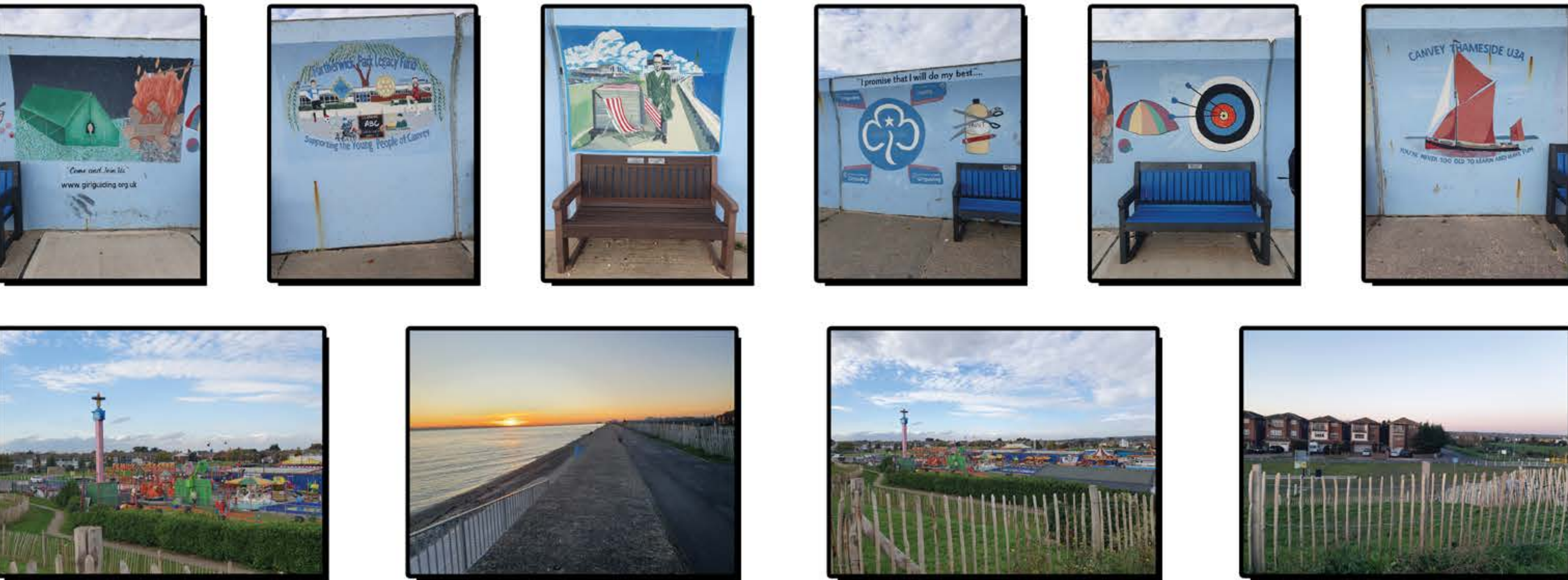
Wall / elevated boundary:



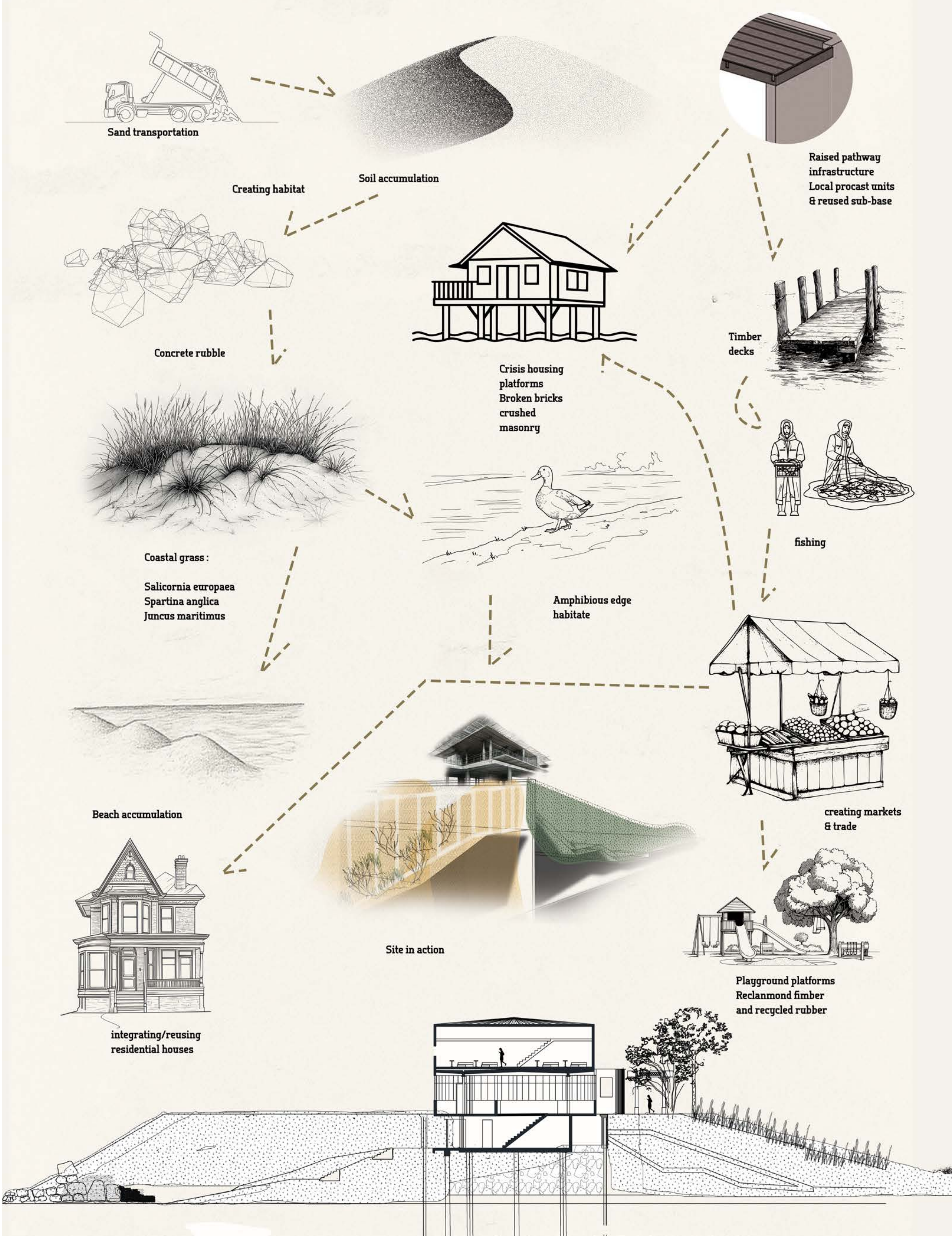
Edge Conditions:



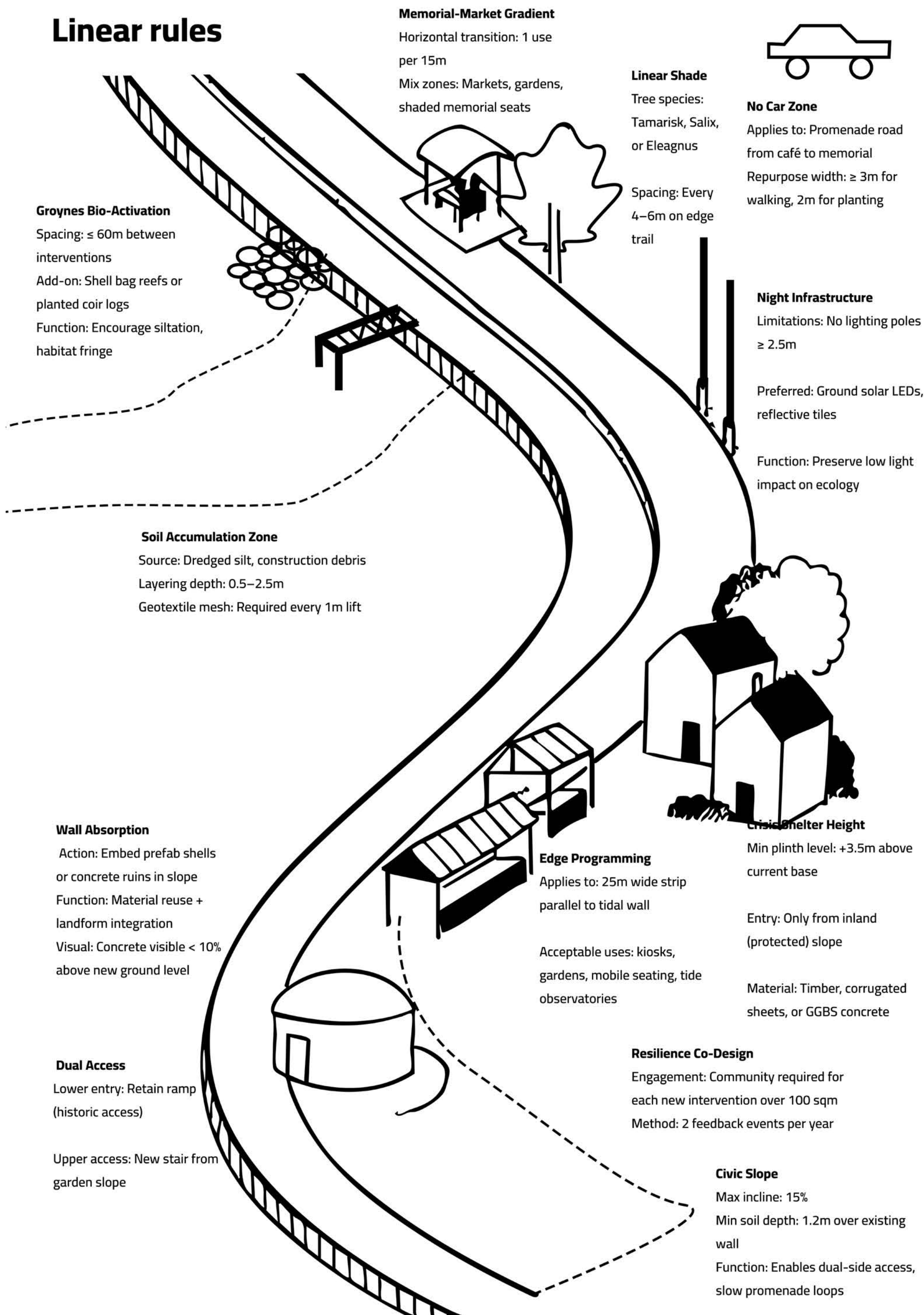
Memory + Social Space:



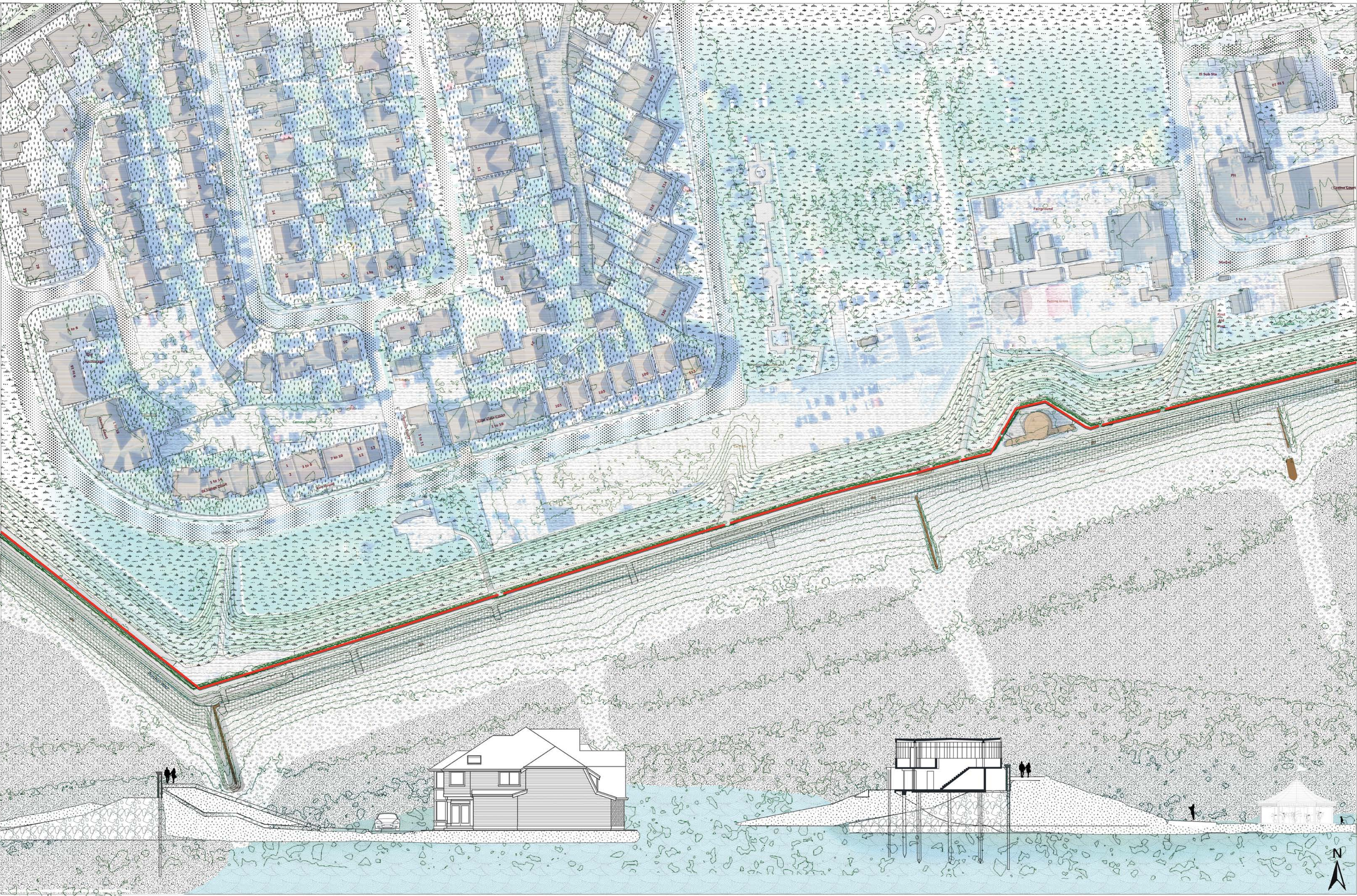
Metabolic Elements: Rising Soils and Site Activities

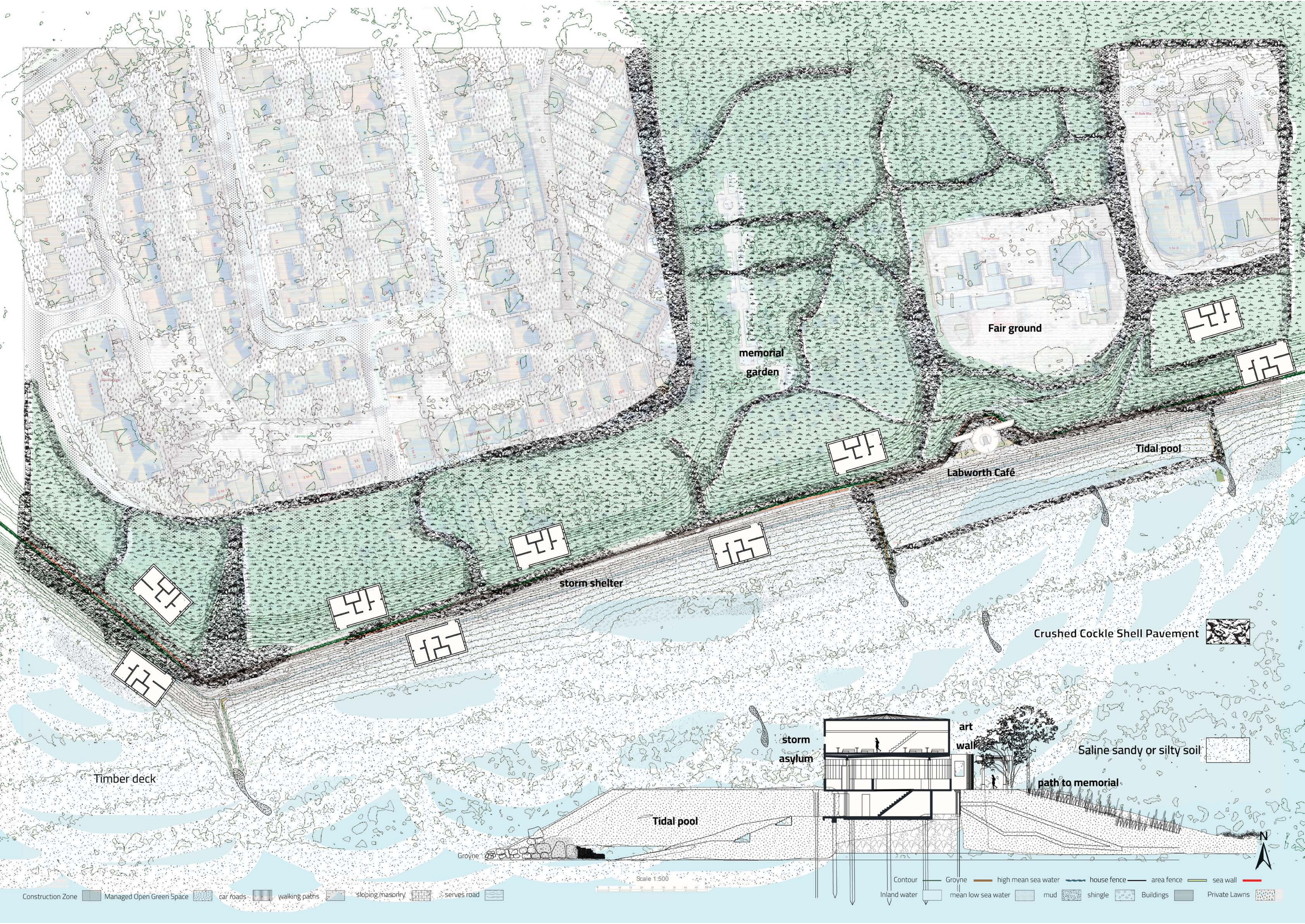


Linear rules



Current Site





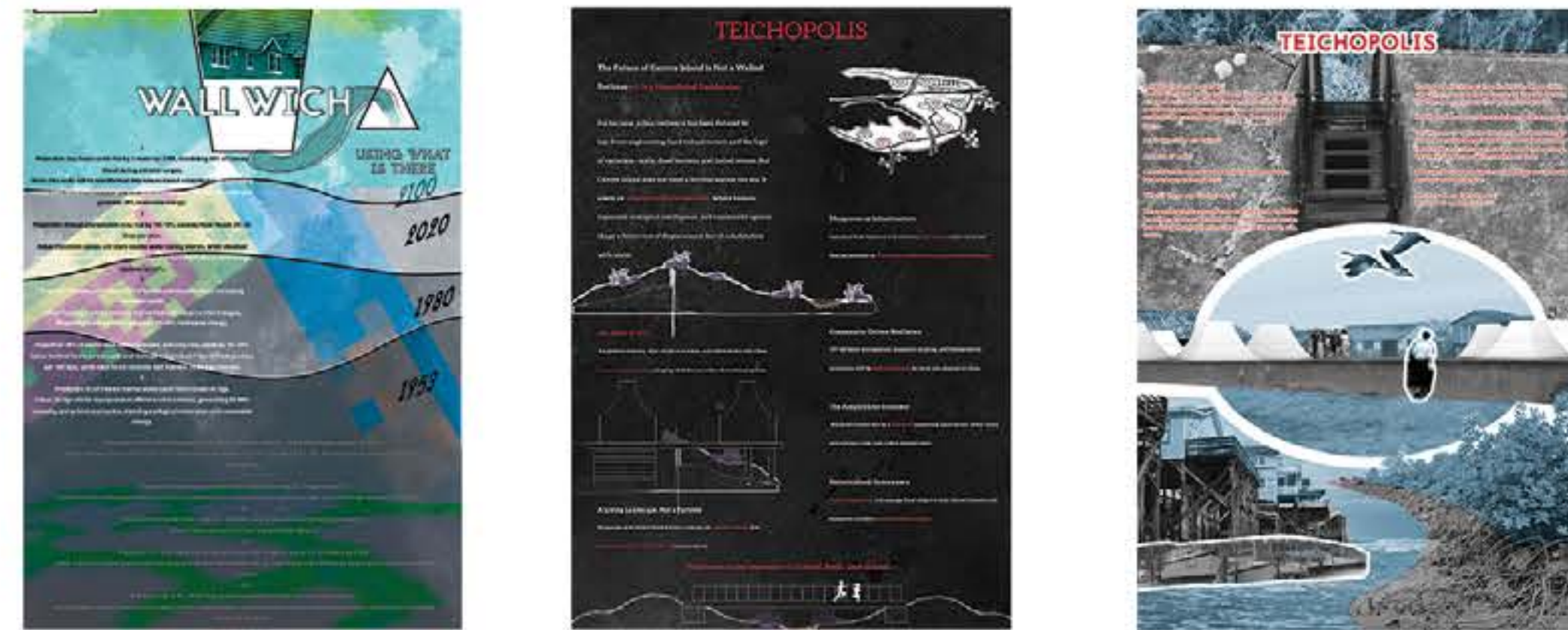


Appendix

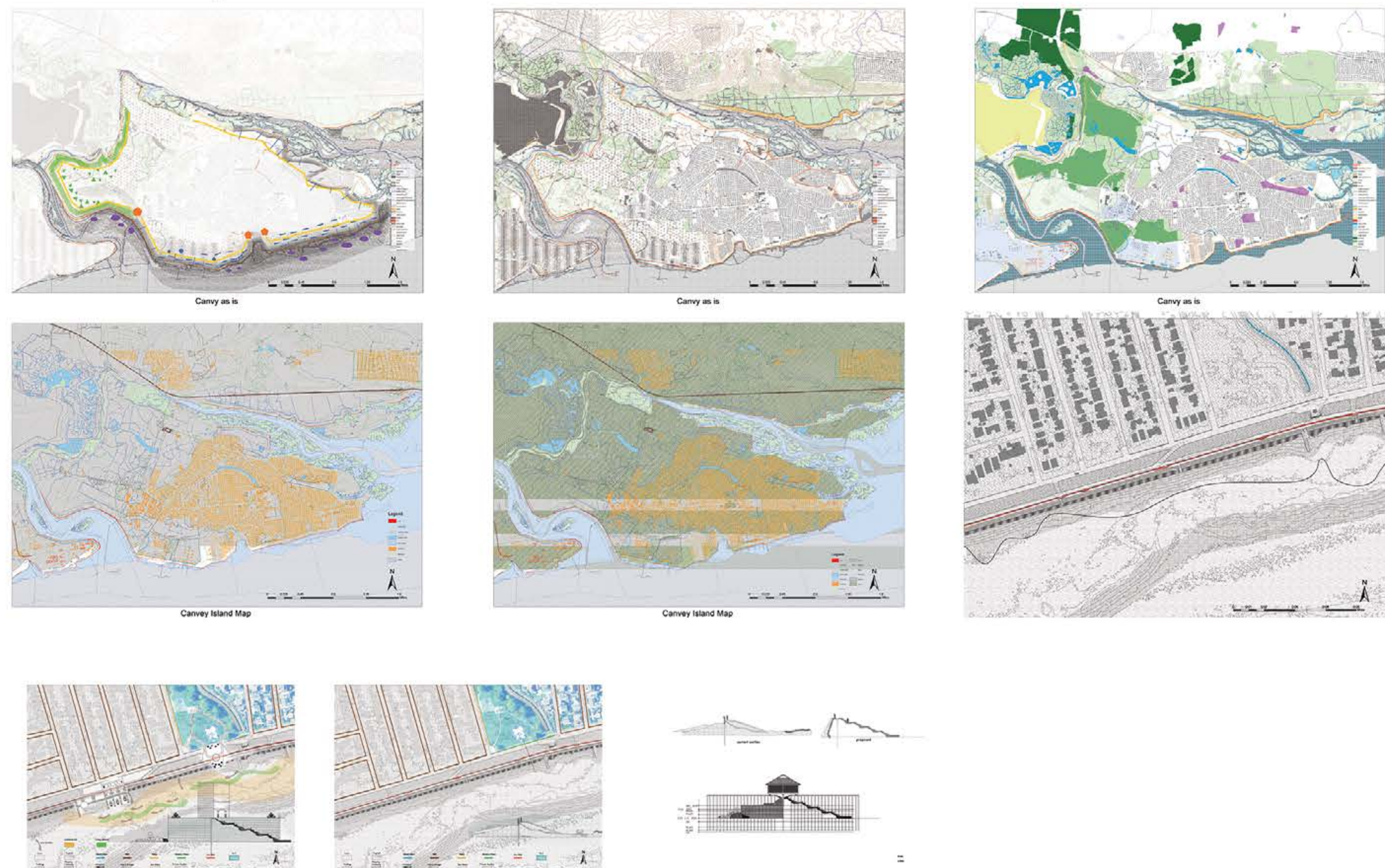
collages



Manifestos



Detail development



sketch

